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Distributed Control Scheme for the Coordination of Interlinking Converters in Islanded Hybrid AC/DC Multi-Microgrids

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ABSTRACT Multiple interconnected ac and dc microgrids (MGs) are being studied by academia and industry because of their benefits despite their operational challenges. Coordinating distributed generators (DGs) is complex, so communication-based controllers are proliferating in the literature. Then, this work proposes a distributed control strategy for islanded ac/dc multi-MGs interconnected by interlinking converters (ILCs). The proposed scheme is implemented in the ILCs and consists of distributed controllers that equalize global generation costs, allowing secondary control in each MG. Control actions that safeguard the saturated operation of MGs and ILCs are included in the control designs. The simultaneous operation with multiple objectives is possible due to adjusting control parameters according to a prioritization criterion. Experiments are conducted through an extensive simulated environment. The results show the proposed multiobjective controllers could maintain global optimal costs during normal operation while not overloading DGs, ILCs, subgrids, and clusters of ILCs. Furthermore, the strategy may reduce operational costs in the long term by protecting the lifetime of critical MG components. Desynchronization of incremental costs is enforced at 15% at most under demanding conditions. Also, it is possible to operate against considerable (>250 [ms]) time delays in the tests.

INDEX TERMS AC/DC microgrid (MG), distributed control, economic dispatch, interlinking converter (ILC), multi-MG, multiobjective control.

NOMENCLATURE

MG	Microgrid.
DG	Distributed generator.
ILC	Interlinking converter.
EMS	Energy management system.
DNO	Distributed network operator.
IC	Incremental cost.

I. INTRODUCTION

Providing flexibility and modularization to future electrical networks is essential to improve the power quality, management of energy resources, and reliability at different points of the system. To this end, the literature has promoted over the last 10 years the use of MGs [1], [2], [3]. In general terms, MGs comprise a group of DGs and loads that can

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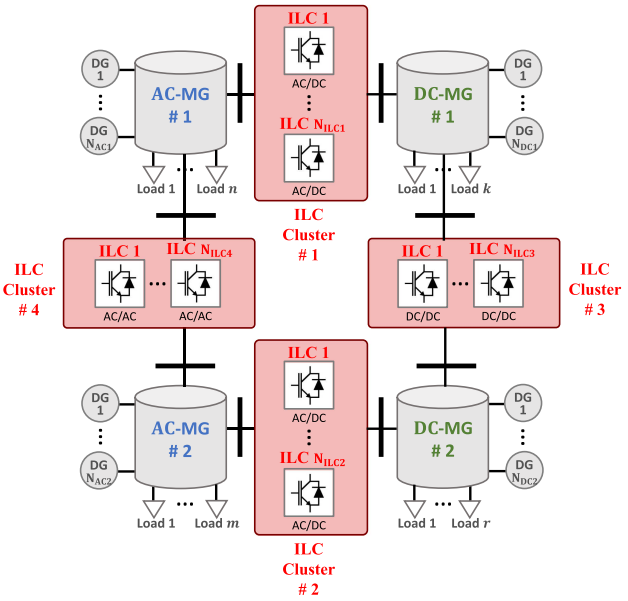

FIGURE 1. Representation of a meshed and hybrid ac/dc multi-MG with four clusters of ILCs.

manage their energy resources autonomously. The modernization of electrical grids through MGs carries information and communication technology integration, particularly deploying autonomous and communicated controllers for the DGs. This favors the evolution towards smart MGs, where an efficient and economical operation is more feasible than conventional grids.

The combination of MGs has caught special interest because this brings even more flexibility to the electrical systems [4]. Therefore, one of the research trends concerning MGs is related to the implemented topologies; in particular, hybrid ac/dc MGs have been extensively studied for real applications since they keep the advantages of the dc systems while reusing most of the existing ac power systems [2], [5]. In this regard, some researchers have proposed to enlarge the hybrid MG systems to multi-MGs [6], [7]. Also, the authors in [8], [9], and [10] have proposed multiple ILCs for the interconnection to increase the total power capacity while improving reliability against unit failures.

In Fig. 1, a generic representation of ac/dc multi-MG is shown. The topology is generalized in this representation as a meshed system, where multiple ILCs are deployed to interconnect the subgrids; other multi-MG systems could be seen as simplifications of the system depicted in Fig. 1.

A. PROBLEM STATEMENT

Within multi-MGs, global control actions, such as power sharing and optimal dispatch, require a certain level of coordination between the subgrids. In this sense, there is a challenge when defining a communication topology to deal with the tradeoff between robustness and investment costs. In addition, as mentioned in [11] and [12], EMSs for economic dispatch must consider restrictions to guarantee a safe and efficient operation. In this regard, local power restrictions specific to each MG must be incorporated into the global coordination of the system.

In the case of multi-MG with multiple ILCs, coordination is required within each ILC cluster so that the system operates properly and there is no unnecessary power flows through the system (causing losses), as previously mentioned in [8], [10], [13], [14], and [15]. Here, it is essential to avoid the unnecessary power flow inside a cluster and between clusters of ILCs. Furthermore, it should be considered to seek to take advantage of the information available by the control strategy so that the DNO can define additional restrictions and deliver an optimal power flow that adjusts to the needs of the distribution multi-MG.

B. MOTIVATION

Some of the advantages of multi-MGs are lower operation costs and higher profits that MG owners (or communities) can receive when compared with single MG systems [7], [16], [17], [18], [19]. Also, multi-MG consumers can benefit from a more reliable and economical power supply [17]. In terms of the ILC construction, special cases are the ILCs for dc–dc and ac–ac MGs. For such cases, small modifications need to be made compared with the control logic employed in ac/dc ILCs (discussions about the ILC construction are outside the scope of this work). It is worth noting that an ac/ac ILC adjusts the MG's voltage and frequency levels; the latter allows the integration of customized industrial subgrids into a global coordination scheme. Moreover, using ILCs for the interconnection of multiple MGs gives freedom for each MG to operate with its own voltage level and/or frequency.

Despite its potential, the use of ac/dc meshed topologies with ILCs (such as the one shown in Fig. 1) is barely covered in the literature (as in [7], [20], [21], and [22]). Consequently, there is a lack of literature on the control design and coordination of such meshed systems.

Until now, meshed topologies with multiple ILCs were considered complex to study, and the available literature only focuses on multiple ac MGs partitioned by isolation switches [7], [21], [23], [24]. The potential contribution in this regard is significant because in complex networks, such as the one that is addressed, centralized control and optimizers entail significant disadvantages in terms of costs, reliability, and time of execution.

C. CONTRIBUTIONS AND ORGANIZATION

This article proposes a cooperative control scheme for the ILCs in a meshed ac/dc multi-MG system with ILCs (that could be dc/dc, ac/ac or ac/dc). The proposed scheme aims to achieve economic dispatch by equalizing the IC variables shared by a communication layer between ILCs and subgrids. The strategy performs additional control goals related to the balancing of power inside ILC clusters, and the balancing of average power between ILC clusters and subgrids to avoid saturations. The control scheme uses adjustable weights based

on exponential activation functions to deal with the tradeoffs between economic and safety operations.

The contributions of this article can be summarized as follows.

1) Optimal dispatch formulation is established considering the following two fundamental aspects that have been insufficiently addressed in the literature:
   a) the topology of the meshed hybrid ac/dc multi-MG system with clusters of ILCs, where the power balance constraint is proposed using a connectivity matrix and nonlinear functions representing the ILC powers as dependent on the power of all the DGs in the system;
   b) economic dispatch with multiple objectives, where terms inside the objective function are added to regularize the avoidance of MG saturation and ILC power sharing inside ILC clusters.
2) A distributed multiobjective control for the ILCs is proposed, which guarantees the economic dispatch without communication between ILC clusters. Also, MG saturation and ILC power sharing are implemented through control loops relying on averaged values of powers from neighbor DGs and other ILCs inside the ILC cluster.
3) A novel constraint for the economic dispatch optimization problem of a meshed hybrid ac/dc multi-MG system is proposed to restrict the power saturation of ILC clusters. This constraint is implemented as a term in a distributed multiobjective controller, which yields the consensus of average powers of ILCs received from neighbor ILC clusters.
4) A case study of meshed hybrid ac/dc multi-MG with clusters of ILCs is simulated and evaluated under different conditions.

The rest of this article is organized as follows. Section II presents the formulation of the economic dispatch problem and distributed control design for the ILC controllers. The design includes multiple control objectives: for IC consensus, MG saturation, and ILC power sharing. In Section III, the formulation and distributed control design of a novel ILC cluster saturation constraint is performed, which is added to the developments of the previous section. Section IV describes the case studies, providing details about the simulated system and control parameters. In Section V, the simulation results are presented and discussed. Finally, Section VI concludes this article.

II. DESIGN OF DISTRIBUTED CONTROL FOR A MESHED MULTI-MG WITH CLUSTERS OF ILCS

This section describes the application of distributed coordination control of ILCs for a generalized meshed multi-MG; this topology incorporates several subgrids of ac and dc nature interconnected by clusters of ILCs.

It is worth noting that the use of multiple parallel converters may incorporate harmonics into the grid. Then, using multiple ILCs in a cluster connected to the same dc MG might bring circulating currents and harmonics [25], which may need to be compensated depending on the system's robustness. Although important, these and other power quality aspects are outside the scope of this work, so they are not part of the following formulations.

A. FORMULATION OF ECONOMIC DISPATCH IN A MESHED AC/DC MULTI-MG

Let $\boldsymbol{P}$ be the vector with the powers of all the DGs in the multi-MG. Also, let $N_{\text{sys}} = N_1^{\text{MG}} + \cdots + N_{N_{\text{MG}}}^{\text{MG}}$ represent the total number of DGs in the system composed by N_{MG} MGs, with N_m^{MG} the number of DGs in the mth MG, and $N_{\text{ILC}} = N_1^{\text{cl}} + \cdots + N_{N_{\text{cl}}}^{\text{cl}}$ represent the number of ILCs in the system composed by N_{cl} ILC clusters, with N_j^{cl} the number of ILCs in the jth ILC cluster.

Then, the economic dispatch optimization can be written as

$$\min_{\boldsymbol{P}} \left\{ \sum_{i=1}^{N_{\text{sys}}} C_i(P_i) \right\} \tag{1}$$

subject to

$$C_i(P_i) = a_{ci}P_i^2 + b_{ci}P_i + c_{ci} \qquad \forall\, i \in \mathcal{N}_{\text{sys}} \tag{1a}$$

$$P_i^{\min} \le P_i \le P_i^{\max} \qquad \forall\, i \in \mathcal{N}_{\text{sys}} \tag{1b}$$

$$-P_{\text{ILC}i}^{\max} \le P_{\text{ILC}i}(\boldsymbol{P}) \le P_{\text{ILC}i}^{\max} \qquad \forall\, i \in \mathcal{N}_{\text{ILC}} \tag{1c}$$

$$P_D^m - \sum_{i=1}^{N_m^{\text{MG}}} P_i = \sum_{j=1}^{N_{\text{cl}}} \phi_{mj} \sum_{l=1}^{N_j^{\text{cl}}} P_{\text{ILC}l}(\boldsymbol{P}) \qquad \forall\, m \in \mathcal{N}_{\text{MG}} \tag{1d}$$

where $C_i(P_i)$ is a quadratic cost function for the ith DG, with parameters a_{ci}, b_{ci}, and c_{ci}. Also, $P_{\text{ILC}i}$ is the power of the ith ILC, and P_{D}^m is the demanded power of the mth MG. It is worth noticing $P_{\text{ILC}}^{\min} = -P_{\text{ILC}}^{\max}$ by ILC's symmetry.

Constraint (1d) represents the power balance of the multi-MG. It has the physical connectivity matrix between ILC clusters and MGs given by

$$\phi = \begin{bmatrix} \phi_{11} & \cdots & \phi_{1N_{\text{cl}}} \\ \vdots & \ddots & \vdots \\ \phi_{N_{\text{MG}}1} & \cdots & \phi_{N_{\text{MG}}N_{\text{cl}}} \end{bmatrix} \tag{2}$$

such that $\phi_{mj} \in \{1, -1\}$ if there is a connection between the mth MG and the jth ILC cluster; otherwise, $\phi_{mj} = 0$. The condition $\mathbf{1}_m^T \phi = 0$ must hold, i.e., there is a conservative power flow between ILC clusters. Thereafter, when summing the constraint equations of (1d), the term with $P_{\text{ILC}l}$ disappears due to the properties of ϕ.

The inequality constraints can be converted into a set of equality constraints, producing a Lagrangian function of the form

$$\mathbb{L}\left(P_i, \sigma_i^+, \sigma_i^-, \Lambda_j^+, \Lambda_j^-, \lambda\right)$$

$$= \sum_{i=1}^{N_{\text{sys}}} C_i(P_i) + \sigma_i^+(P_i - P_i^{\max}) + \sigma_i^-(P_i^{\min} - P_i)$$

$$+\sum_{j=1}^{N_{\mathrm{ILC}}}\Lambda_j^+\left(P_{\mathrm{ILC}j}(\boldsymbol{P})-P_{\mathrm{ILC}j}^{\max}\right)+\Lambda_j^-\left(-P_{\mathrm{ILC}j}^{\max}-P_{\mathrm{ILC}j}(\boldsymbol{P})\right)$$

$$+\lambda\left(\sum_{m=1}^{N_{\mathrm{MG}}}P_{\mathrm{D}}^m-\sum_{i=1}^{N_{\mathrm{sys}}}P_i\right) \tag{3}$$

where λ, σ_i^+, σ_i^-, Λ_j^+, and Λ_j^- are Lagrange multipliers. The multipliers σ_i^+ and σ_i^- are defined by a control loop, as in [26], to limit the output of P_i as follows:

$$\dot{\sigma}_i^+=\frac{k_3}{k_1}\max\left(0\,,\ P_i-P_i^{\max}+\frac{k_2}{k_3}\sigma_i^+\right)-\frac{k_2}{k_1}\sigma_i^+ \tag{4}$$

$$\dot{\sigma}_i^-=\frac{k_3}{k_1}\max\left(0\,,\ P_i^{\min}-P_i+\frac{k_2}{k_3}\sigma_i^-\right)-\frac{k_2}{k_1}\sigma_i^- \tag{5}$$

where k_1, k_2, and k_3 are control parameters.

Similarly, multipliers Λ_j^+ and Λ_j^- are defined, analogously to σ_i^+ and σ_i^-, to restrict power violations on the jth ILC.

With those definitions, the stationary optimality condition can be checked via the derivative of the Lagrangian function as follows:

$$\frac{\partial}{\partial P_i}\mathbb{L}=\frac{\partial C_i\left(P_i\right)}{\partial P_i}+\sum_{j=1}^{N_{\mathrm{ILC}}}\left(\Lambda_j^+-\Lambda_j^-\right)\frac{\partial P_{\mathrm{ILC}j}(\boldsymbol{P})}{\partial P_i}+\sigma_i^+-\sigma_i^--\lambda=0\,. \tag{6}$$

Then, one has

$$\lambda=\frac{\partial C_i\left(P_i\right)}{\partial P_i}+\sigma_i^+-\sigma_i^-+\sum_{j=1}^{N_{\mathrm{ILC}}}\left(\Lambda_j^+-\Lambda_j^-\right)\frac{\partial P_{\mathrm{ILC}j}(\boldsymbol{P})}{\partial P_i} \tag{7}$$

which is equivalent to what was derived in [11] and [26] if there are no power violations on the ILCs ($\Lambda_j^+-\Lambda_j^-=0$). Then, λ represents the global IC dynamics.

Remark 1: It is worth noting that the ICs of the side MGs can synchronize even if there are multipliers Λ_j^+ or Λ_j^- different to zero; this is true if and only if there is, at least, one ILC that is not saturated, i.e., $\prod_{j=1}^{N_{\mathrm{ILC}}}\Lambda_j^+\Lambda_j^-=0$.

Remark 2: The optimization in (1) considers the global economic dispatch, i.e., all the DGs cooperate to achieve the cheapest operation. Other aspects such as reducing power losses or pollution emissions are not considered in this formulation, although they can be performed by the DNO through power references, as will be mentioned in Section III-B.

Remark 3: All of the DG cost functions $C_i(P_i)$ can be assumed using the same form, i.e., a quadratic function, since it generalizes the majority of energy sources' operational costs. However, other cost functions could be adopted for renewable energy generators, such as linear [27] or piecewise forms [28], without altering the distributed methodology that will be described in the next sections.

Remark 4: The term $\frac{\partial P_{\mathrm{ILC}j}(\boldsymbol{P})}{\partial P_i}$ is a linear combination of the powers in $\boldsymbol{P}$, which is feasible to obtain provided full knowledge of the multi-MG electrical topology, including the physical interconnections of DGs and ILC clusters. Its use in this work is to represent the deviations of global IC, λ, when there are violations of the defined system's operational constraints.

B. FORMULATION OF ECONOMIC DISPATCH IN A MESHED AC/DC MULTI-MG WITH ILC POWER SHARING, HARD POWER CONSTRAINTS AND SUBGRID POWER SATURATION

Based on the advancements of [11] and [14], this section presents a formulation for the global economic dispatch of multiple interconnected MGs with additional control goals. In particular, the regulation of MG saturations and the ILC power balancing are studied along with hard constraints of power in some of the ILCs.

Then, from (1), an expanded multiobjective global dispatch optimization can be expressed as

$$\min_{\boldsymbol{P}}\left\{J\left(f_1(\boldsymbol{P}),\,f_2(\boldsymbol{P}),\,f_3(\boldsymbol{P})\right)\right\} \tag{8}$$

subject to

$$f_1(\boldsymbol{P})=\sum_{i=1}^{N_{\mathrm{sys}}}C_i(P_i) \tag{8a}$$

$$f_2(\boldsymbol{P})=\sum_{j=1}^{N_{\mathrm{cl}}}\sum_{\substack{m=1\\n=1\\m\neq n}}^{N_{\mathrm{MG}}}\left(\phi_{mj}h_m\overline{P}_m^{\mathrm{MG}}(\boldsymbol{P})+\phi_{nj}h_n\overline{P}_n^{\mathrm{MG}}(\boldsymbol{P})\right) \tag{8b}$$

$$f_3(\boldsymbol{P})=\sum_{j=1}^{N_{\mathrm{cl}}}\sum_{l=1}^{N_j^{\mathrm{cl}}}\left(\overline{P}_j^{cl}(\boldsymbol{P})-\frac{P_{\mathrm{ILC}l}(\boldsymbol{P})}{P_{\mathrm{ILC}l}^{\max}}\right) \tag{8c}$$

$$C_i(P_i)=a_{ci}P_i^2+b_{ci}P_i+c_{ci}\qquad\forall\, i\in\mathcal{N}_{\mathrm{sys}} \tag{8d}$$

$$P_i^{\min}\ \leq\ P_i\ \leq P_i^{\max}\qquad\forall\, i\in\mathcal{N}_{\mathrm{sys}} \tag{8e}$$

$$-P_{\mathrm{ILC}i}^{\max}\leq\ P_{\mathrm{ILC}i}(\boldsymbol{P})\ \leq P_{\mathrm{ILC}i}^{\max}\qquad\forall\, i\in\mathcal{N}_{\mathrm{ILC}} \tag{8f}$$

$$\sum_{m=1}^{N_{\mathrm{MG}}}P_{\mathrm{D}}^m-\sum_{i=1}^{N_{\mathrm{sys}}}P_i=0\qquad\forall\, i\in\mathcal{N}_{\mathrm{sys}},\ m\in\mathcal{N}_{\mathrm{MG}} \tag{8g}$$

$$P_{\mathrm{ILC}k}(\boldsymbol{P})-P_{\mathrm{ILC}k}^{\mathrm{ref}}=0\qquad\forall\, k\in\mathcal{N}_k\subseteq\mathcal{N}_{\mathrm{ILC}}\,. \tag{8h}$$

The objective function $f_1(\boldsymbol{P})$ represents the original economic dispatch. The objective functions $f_2(\boldsymbol{P})$ and $f_3(\boldsymbol{P})$ represent the additional objectives, which are the balancing among the power utilization of MGs [11] and among ILCs in a cluster [14], respectively. The functions $\overline{P}_m^{\mathrm{MG}}(\boldsymbol{P})$ and $\overline{P}_j^{cl}(\boldsymbol{P})$ are the per unit average power of the mth MG and jth ILC cluster, respectively. Coefficients $h_m\,\forall\, m\in\mathcal{N}_{\mathrm{MG}}$ determine the prioritization in the management of saturation/power reserves of MGs. Also, as described in [14], the system can be generalized by including power reference commands into the ILCs; these power references could be assumed generated by a DNO. Also, N_k is the number of ILCs with power references arbitrarily selected by the DNO.

In order to solve (8), assumptions can be made in the objective functions. One way to deal with the simultaneous optimizations is to regularize with weighting parameters. These changes give rise to the minimization of the objective junction

$$\min_{\boldsymbol{P}} \{f_1(\boldsymbol{P}) + hf_2(\boldsymbol{P}) + wf_3(\boldsymbol{P})\} \tag{9}$$

where h and w are weighting parameters regulating the trade-off between control objectives.

Following the steps of the previous section, one has that the derivative of the Lagrangian function is

$$\frac{\partial}{\partial P_i}\mathbb{L} = \frac{\partial C_i\,(P_i)}{\partial P_i} + hg_1(\boldsymbol{P}) + wg_2(\boldsymbol{P}) + \sigma_i^+ - \sigma_i^- - \lambda + \sum_{j=1}^{N_{\text{ILC}}}(\Lambda_j^+ - \Lambda_j^-)\frac{\partial P_{\text{ILC}j}(\boldsymbol{P})}{\partial P_i} + \sum_{k=1}^{N_k} z_k \frac{\partial P_{\text{ILC}k}(\boldsymbol{P})}{\partial P_i} \tag{10}$$

where z_k are the Lagrange multipliers associated with the hard constraints of power, and $g_1(\boldsymbol{P})$ and $g_2(\boldsymbol{P})$ are linear functions, which can be approximated by

$$g_1(\boldsymbol{P}) = \frac{\partial f_2(\boldsymbol{P})}{\partial P_i} \approx \sum_{j=1}^{N_{\text{cl}}} \frac{\phi_{xj} h_x}{N_x^{\text{MG}}}$$

$$g_2(\boldsymbol{P}) = \frac{\partial f_3(\boldsymbol{P})}{\partial P_i} \approx \sum_{j=1}^{N_{\text{cl}}}\sum_{l=1}^{N_j^{\text{cl}}}\left(\sum_{k=1}^{N_j^{\text{cl}}} \frac{1}{N_j^{\text{cl}} P_{\text{ILC}k}^{\max}} \frac{\partial P_{\text{ILC}k}(\boldsymbol{P})}{\partial P_i} - \frac{1}{P_{\text{ILC}l}^{\max}} \frac{\partial P_{\text{ILC}l}(\boldsymbol{P})}{\partial P_i}\right) \tag{11}$$

where $x \in \mathcal{N}_{\text{MG}}$ such that $i \in \mathcal{N}_x^{\text{MG}}$.

With such a system, the stationary optimality condition gives

$$\lambda = \frac{\partial C_i\,(P_i)}{\partial P_i} + hg_1(\boldsymbol{P}) + wg_2(\boldsymbol{P}) + \sigma_i^+ - \sigma_i^- + \sum_{j=1}^{N_{\text{ILC}}}(\Lambda_j^+ - \Lambda_j^-)\frac{\partial P_{\text{ILC}j}(\boldsymbol{P})}{\partial P_i} + \sum_{k=1}^{N_k} z_k \frac{\partial P_{\text{ILC}k}(\boldsymbol{P})}{\partial P_i}\,. \tag{12}$$

Remark 5: Noting that $\lambda_l = \lambda_k\ \forall\, l, k \in \mathcal{N}_{\text{sys}}$ must be held to achieve the cheapest operation cost. If $\{h, w\} = \mathbf{0}$, the ICs between all the DGs in the ac/dc multi-MG are synchronized using (12) provided there is at least one ILC without saturation, i.e., $\prod_{j=1}^{N_{\text{cl}}}\prod_{l=1}^{N_j^{\text{cl}}} \Lambda_l^+\Lambda_l^- = 0$. Also, there must be at least one ILC cluster without a single hard constraint of power in it.

From (12), one can see terms that deviate the optimal solution of the conventional economic dispatch. In particular, it is possible to design the weighting parameters h and w such that the system deviates the optimum IC regarding some safety operational criteria, such as having energy reserves for an MG failure. Hence, a distributed control scheme can be realized to solve the multiobjective optimization problem by incorporating weighting parameters in each ILC's control loop.

C. COMMUNICATION NETWORK OF A HYBRID MULTI-MG WITH MULTIPLE CLUSTERS OF ILCS

Concerning the communication topology, the multi-MG with multiple clusters of ILCs can be represented by an equivalent graph $\mathcal{G}_{\text{sys}} := \cup_{i=1}^{N_{\text{MG}}} \mathcal{G}_{\text{MG},i} \cup_{j=1}^{N_{\text{cl}}} \mathcal{G}_{\text{cl},j}$, where $\mathcal{G}_{\text{MG},i}$ is the graph of the ith MG and $\mathcal{G}_{\text{cl},j}$ is the graph of the jth ILC cluster; each graph can operate independently in case of cluster failures. In addition, $\mathcal{G}_{\text{MG},i} := (\mathcal{N}_i^{\text{MG}}, E_i^{\text{MG}}, A^i)$ with $\mathcal{N}_i^{\text{MG}}$ the set of nodes, E_i^{MG} the set of edges and A^i the adjacency (communication) matrix. Similarly, $\mathcal{G}_{\text{cl},j} := (\mathcal{N}_j^*, E_j^{\text{cl}}, A^{\text{cl},j})$ with $\mathcal{N}_j^* \subset (\mathcal{N}_m^{\text{MG}} \cup \mathcal{N}_n^{\text{MG}} \cup \mathcal{N}_j^{\text{cl}})$ for any pair of MGs, m and n, connected by the jth ILC cluster, E_j^{cl} is the associated set of edges, and $\{A^{\text{cl},j}\} = \cup_{i=1}^{N_j^{\text{cl}}}\{\boldsymbol{a}_m^{\text{ILC}i}\} \cup_{i=1}^{N_j^{\text{cl}}}\{\boldsymbol{a}_n^{\text{ILC}i}\} \cup_{i=1}^{N_j^{\text{cl}}}\{\boldsymbol{a}_{\text{ILC}}^{\text{ILC}i}\}$ where $\boldsymbol{a}_m^{\text{ILC}i}$, $\boldsymbol{a}_n^{\text{ILC}i}$, and $\boldsymbol{a}_{\text{ILC}}^{\text{ILC}i}$ are vectors that represent the communication between an ILC and the DGs of the mth and nth MGs, and with the other ILCs in the jth cluster, respectively.

In the matrix form, the communication network of the ac/dc multi-MG system can be represented by

$$A^{\text{sys}} = \begin{bmatrix} A^{\text{MMG}} & A_{\text{MMG}}^{\text{ILCs}} \\ A_{\text{MMG}}^{\text{ILCs}} & A_{\text{ILCs}}^{\text{ILCs}} \end{bmatrix} \tag{13}$$

$$A^{\text{MMG}} = \begin{bmatrix} A^m & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & A^{N_{\text{MG}}} \end{bmatrix} \tag{14}$$

$$A_{\text{MMG}}^{\text{ILCs}} = \begin{bmatrix} A_{\text{MG},1}^{\text{cl},1} & \cdots & A_{\text{MG},N_{\text{MG}}}^{\text{cl},1} \\ \vdots & \vdots & \vdots \\ A_{\text{MG},1}^{\text{cl},N_{\text{cl}}} & \cdots & A_{\text{MG},N_{\text{MG}}}^{\text{cl},N_{\text{cl}}} \end{bmatrix} \tag{15}$$

$$A_{\text{ILCs}}^{\text{ILCs}} = \begin{bmatrix} A_{\text{cl},1}^{\text{cl},1} & \cdots & A_{\text{cl},N_{\text{cl}}}^{\text{cl},1} \\ \vdots & \vdots & \vdots \\ A_{\text{cl},1}^{\text{cl},N_{\text{cl}}} & \cdots & A_{\text{cl},N_{\text{cl}}}^{\text{cl},N_{\text{cl}}} \end{bmatrix} \tag{16}$$

where A^m is the adjacency matrix of the mth MG. A^{MMG}, $A_{\text{MMG}}^{\text{ILCs}}$, and $A_{\text{ILCs}}^{\text{ILCs}}$ are the communication matrices describing the communications between DGs, DGs and ILCs, and between ILCs, respectively. Moreover

$$A_{\text{MG},m}^{\text{cl},j} = \left[\boldsymbol{a}_m^{\text{ILC}1} \cdots \boldsymbol{a}_m^{\text{ILC}N_j^{\text{cl}}}\right]^{\text{T}} \tag{17}$$

$$A_{\text{cl},j}^{\text{cl},k} = \left[\boldsymbol{a}_{\text{cl},j}^{\text{ILC}1} \cdots \boldsymbol{a}_{\text{cl},j}^{\text{ILC}N_k^{\text{cl}}}\right]^{\text{T}} \tag{18}$$

$$A_{\text{cl},j}^{\text{cl},j} = A_{\text{ILC}}^{\text{cl},j} = \left[\boldsymbol{a}_{\text{ILC}}^{\text{ILC}1} \cdots \boldsymbol{a}_{\text{ILC}}^{\text{ILC}N_j^{\text{cl}}}\right]^{\text{T}} \tag{19}$$

with the vectors $\boldsymbol{a}_m^{\mathrm{ILC}i} = (a_1^{\mathrm{ILC}i}, \ldots, a_{N_m^{\mathrm{MG}}}^{\mathrm{ILC}i})$ and $\boldsymbol{a}_{\mathrm{cl},j}^{\mathrm{ILC}i} = (a_{\mathrm{ILC}1}^{\mathrm{ILC}i}, \ldots, a_{\mathrm{ILC}N_j^{\mathrm{cl}}}^{\mathrm{ILC}i})$ representing the communication of the ith ILC with the DGs in the m-MG, and with the ILCs of the jth ILC cluster, respectively. Also, $\boldsymbol{a}_{\mathrm{ILC}}^{\mathrm{ILC}i} = (a_{\mathrm{ILC}1}^{\mathrm{ILC}i}, \ldots, a_{\mathrm{ILC}N_j^{\mathrm{cl}}}^{\mathrm{ILC}i})$ represents the special case of the i-th ILC communicating with other ILCs inside its cluster.

The form of A^{MMG} suggests that no communication between DGs of different MGs is allowed. Also, in this section, the ILCs of different clusters are assumed not communicated, i.e., $A_{\mathrm{cl},j}^{\mathrm{cl},i} = 0 \; \forall i \neq j \; \wedge \; i, j \in \mathcal{N}_{\mathrm{cl}}$.

The former definitions are used to design distributed controllers in the DGs and ILCs of the ac/dc multi-MG.

D. DISTRIBUTED CONTROL FOR DGS USING CONSENSUS PROTOCOLS

The distributed control for DGs can be chosen based on already published methods, such as [26], [29], [30], [31], and [32]. In particular, based on [33] and [34], the authors of this article implemented a variant to achieve voltage and frequency restoration while an IC consensus is performed in finite time.

For a generic dc MG "n," the structure of the ith DG's secondary control can be defined in the Laplace domain as

$$E_i^{\mathrm{ref}} = E_{\mathrm{DC}}^* - m^{\mathrm{DC}} P_i + \delta E_i^1 + \delta E_i^2 \tag{20}$$

$$\delta E_i^1 = k_p^E \left(E_{\mathrm{DC}}^* - \overline{E}_i\right) + \frac{k_i^E}{s}\left(E_{\mathrm{DC}}^* - \overline{E}_i\right) \tag{21a}$$

$$\overline{E}_i = E_i + \frac{1}{s}\sum_{j=1}^{N_n^{\mathrm{MG}}} a_{ij}^n \; \mathrm{sig}\left[\overline{E}_j - \overline{E}_i\right]^{\nu^1} \tag{21b}$$

$$\delta E_i^2 = k_p^{\lambda}(u_i^{\lambda}) + \frac{k_i^{\lambda}}{s} u_i^{\lambda} \tag{22a}$$

$$u_i^{\lambda} = c_{\lambda}^{\mathrm{DC}} \; \mathrm{sig}\left[\sum_{j=1}^{N_n^{\mathrm{MG}}} a_{ij}^n \left(\lambda_j - \lambda_i\right)\right]^{\nu^2} \tag{22b}$$

where E_i and P_i are the output voltage and power of the ith DG in the dc MG, E_{DC}^* is the MG reference voltage, m^{DC} is the droop gain, $\{k_p^E, k_i^E, k_p^{\lambda}, k_i^{\lambda}\} > 0$ are parameters of PI controllers, $c_{\lambda}^{\mathrm{DC}} > 0$ is a convergence speed parameter, and parameters $\{\nu^1, \nu^2\} \in (0, 1)$ for $\mathrm{sig}[x]^{\nu} := (x/||x||)||x||^{\nu}$ such that ν regulates the finite-time convergence [35]. The control scheme is resumed in Fig. 2, where $G_{\mathrm{DC}}^E(s)$ and $G_c^{\lambda}(s)$ are the PI controllers.

For a generic ac MG "m," the DGs can adopt the following secondary control loops:

$$E_{di}^{\mathrm{ref}} = E_{\mathrm{AC}}^* - n^{\mathrm{AC}} Q_i + n_d^{\mathrm{AC}} \dot{Q}_i + \delta E_i^1 + \delta E_i^2 \tag{23}$$

$$\delta E_i^1 = \left(k_p^E + \frac{k_i^E}{s}\right)\left(E_{\mathrm{AC}}^* - \overline{E}_i\right) \tag{24a}$$

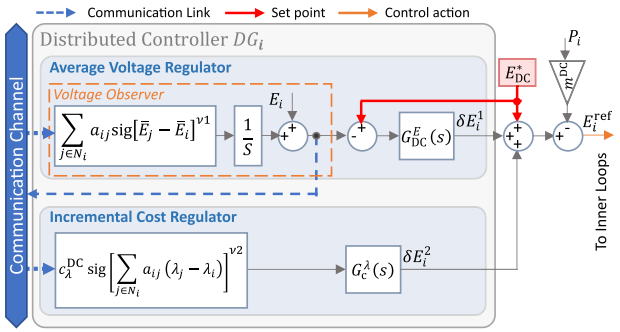

FIGURE 2. Distributed controller for DGs in a dc MG.

$$\overline{E}_i = E_{di} + \frac{1}{s}\sum_{j=1}^{N_m^{\mathrm{MG}}} a_{ij}^m \; \mathrm{sig}\left[\overline{E}_j - \overline{E}_i\right]^{\mu^1} \tag{24b}$$

$$\delta E_i^2 = \left(k_p^Q + \frac{k_i^Q}{s}\right) u_i^Q \tag{25a}$$

$$u_i^Q = c_Q \sum_{j=1}^{N_m^{\mathrm{MG}}} a_{ij}^m \left(\frac{Q_j}{Q_j^{\max}} - \frac{Q_i}{Q_i^{\max}}\right) \tag{25b}$$

$$\omega_i = \omega^* - m^{\mathrm{AC}} P_i + m_d^{\mathrm{AC}} \dot{P}_i + \delta\omega_i^1 + \delta\omega_i^2 \tag{26}$$

$$\delta\omega_i^1 = \frac{c_{\omega}}{s}\left((\omega^* - \omega_i) + \sum_{j=1}^{N_m^{\mathrm{MG}}} a_{ij}^m \left(\delta\omega_j^1 - \delta\omega_i^1\right)\right) \tag{27}$$

$$\delta\omega_i^2 = \frac{c_{\lambda}^{\mathrm{AC}}}{s}\mathrm{sig}\left[\sum_{j=1}^{N_m^{\mathrm{MG}}} a_{ij}^m \left(\lambda_j - \lambda_i\right)\right]^{\mu^2} \tag{28}$$

where E_{di} is the voltage in the direct axis of a d–q reference frame, and $\overline{E}_i$, ω_i, P_i, and Q_i are the local average voltage, frequency, active, and reactive powers of the ith DG in the ac MG, respectively. E_{AC}^* is the MG's voltage reference whereas ω^* is the MG's frequency reference. n^{AC} and m^{AC} are the voltage and frequency droop gains. n_d^{AC} and m_d^{AC} are damping factors to improve transient dynamics. Also, $k_p^E > 0$ and $k_i^E > 0$ are the parameters of a PI voltage controller, $k_p^Q > 0$ and $k_i^Q > 0$ are the parameters of a PI controller whose input is u_i^Q. Moreover, $\{\mu^1, \mu^2\} \in (0, 1)$ are finite-time fractional exponents, and $\{c_Q, c_{\omega}, c_{\lambda}^{\mathrm{AC}}\} > 0$ are convergence speed gains. The control scheme is resumed in Fig. 3, where $G_{\mathrm{AC}}^E(s)$ and $G_c^Q(s)$ are PI controllers.

In steady-state, (21) and (24) achieve an average voltage value for the MG equal to the reference voltage. This allows other consensus protocols to distribute the power between DGs without compromising voltage stability. The asymptotic convergence proof of (21) and (22) is analogous to [31], whereas the finite-time version is to [34]. Similarly, the asymptotic convergence proof of (24)–(28) was demonstrated in [36], [37], and [32], whereas the finite-time convergence in [33] and [38].

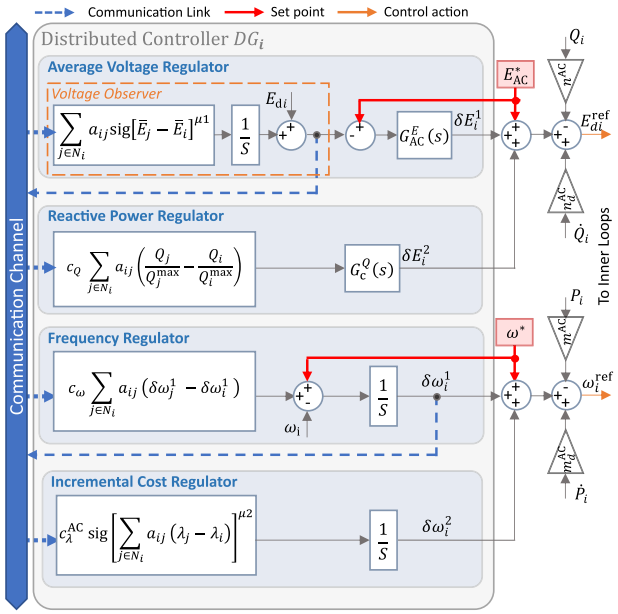

FIGURE 3. Distributed controller for DGs in an ac MG.

Notice that the convergence advantage of finite-time algorithms in secondary control will be small compared with a proper selection of the ILC controllers. A similar case applies for the damping terms n_d^{AC} and m_d^{AC} since they only improve transient dynamics of primary control and can be omitted if there is an appropriate virtual impedance loop or similar technique to decouple between active and reactive power. Therefore, the significant long-term economic savings will come from the designs of ILC controllers in the next sections.

E. DISTRIBUTED MULTIOBJECTIVE CONTROL FOR ILCS USING CONSENSUS PROTOCOL

The distributed control for solving (9) is designed as an extension of [11]. For the ith ILC in the jth ILC cluster, which interconnects the mth MG with the nth MG of a multi-MG system, the ILC control loop takes the form

$$P^*_{\text{ILC}i} = (u_{\text{L}i} + u_{\text{C}i} + u_{\text{P}i} + u_{\text{R}i})G^P_{\text{PI}}(s) \tag{29}$$

$$u_{\text{L}i} = c_L \sum_{l=1}^{N_m^{\text{MG}}} \sum_{k=1}^{N_n^{\text{MG}}} \text{sig}\left[\frac{a_l^{\text{ILC}i}}{N_m^{\text{ILC}}}\lambda_l - \frac{a_k^{\text{ILC}i}}{N_n^{\text{ILC}}}\lambda_k\right]^{\alpha_{\text{L}}} \tag{30}$$

$$u_{\text{C}i} = c_{\text{C}}\left(\overline{P}^{\text{cl}}_{j,i} - \frac{P_{\text{ILC}i}}{P^{\max}_{\text{ILC}i}}\right) \tag{31a}$$

$$\overline{P}^{\text{cl}}_{j,i} = \frac{P_{\text{ILC}i}}{P^{\max}_{\text{ILC}i}} + \frac{c'_{\text{C}}}{s}\sum_{k=1}^{N_j^{\text{cl}}} \text{sig}\left[a^{\text{ILC}i}_{\text{ILC}k}(\overline{P}^{\text{cl}}_{j,k} - \overline{P}^{\text{cl}}_{j,i})\right]^{\alpha_{\text{C}}} \tag{31b}$$

$$u_{\text{P}i} = c_P \sum_{l=1}^{N_m^{\text{MG}}} \sum_{k=1}^{N_n^{\text{MG}}} \text{sig}\left[\frac{a_l^{\text{ILC}i}}{N_m^{\text{ILC}}}h_m\overline{P}^{\text{MG}}_{m,l} - \frac{a_k^{\text{ILC}k}}{N_n^{\text{ILC}}}h_n\overline{P}^{\text{MG}}_{n,k}\right]^{\alpha_{\text{P}}} \tag{32a}$$

$$\overline{P}^{\text{MG}}_{\text{x},l} = \frac{P_l}{P_l^{\max}} + \frac{1}{s}\sum_{k=1}^{N_{\text{x}}^{\text{MG}}} a^{\text{x}}_{lk}(\overline{P}^{\text{MG}}_{\text{x},k} - \overline{P}^{\text{MG}}_{\text{x},l}) \tag{32b}$$

$$u_{\text{R}i} = c_{Ri}\left(P^{\text{ref}}_{\text{ILC}i} - \frac{P_{\text{ILC}i}}{P^{\max}_{\text{ILC}i}}\right) \tag{33}$$

where $N_m^{\text{ILC}} = \boldsymbol{a}_m^{\text{ILC}i} \cdot \mathbf{1}_{N_m^{\text{MG}}}$ is the number of DGs of the mth MG communicating with the ith ILC (active nodes), provided a binary communication matrix. The term $\overline{P}_{\text{x},l}$ is the estimated average power of MG "x" given by its lth DG, and a^{x}_{lk} is the communication weight between DG l and DG k, where $a^{\text{x}}_{lk} \subseteq A^{\text{x}}$ (x could be the MG m or n interfaced by the ith ILC). The terms h_m and h_n are weights that adjust the tradeoff between IC balancing and the subgrids power utilization. Those coefficients could be designed fixed or dependent on power, as the h-function of [11]. In this article, we analyze both cases.

Remark 6: Finite-time protocols can be removed if necessary from (21), (22), (24), (28), (30), (31), and (32) by selecting the exponential parameters equal to 1.

It is worth noting that due to the distributed nature of the proposed consensus protocols, there is no need for knowledge about the MG topologies and multi-MG interconnections. In terms of the communication infrastructure, during the operation, each DG and ILC should know which neighbors are communicating data. Real-time detection schemes can be assumed to be implemented in the communication protocols to identify when a neighbor unit loses communication.

An example of the proposed controller for the ith ILC interconnecting an ac MG with a dc MG is presented in Fig. 4. The representation of the ILC is based on [11].

F. GUIDELINE FOR THE SELECTION OF CONSENSUS PARAMETERS

For the control of secondary control, convergence parameters c_λ^{DC} and c_Q are closely related to the associated PI controllers; specifically, they influence the proportional gain. Then, it is recommended to start from small values to gradually increment the speed, e.g., $0.1 \rightarrow 10$. For the convergence parameters c_λ^{AC} and c_ω, they are gains that regulate the speed of the control loops of IC and frequency, respectively. Attention should be taken to these gains since, for some intervals of values, they may interfere with other control loops, e.g., coupling between IC and reactive power regulation in an ac DG, producing oscillations and instability. The recommendation is to start with small values with gradual increments until the desired control speed is achieved (e.g., with a 3-s settling time), avoiding the proximity with bandwidths of other control loops.

From the previous, the convergence parameters of the ILC controllers can be defined. The IC control speed should not be faster than the IC dynamics of secondary control. Since the ILC power is regulated by a PI controller, c_{L} includes a conversion gain from IC into power, which produces a high value. The tuning of c_{L} shall start with small values with escalations by powers of ten. The parameter c_{P} in this formulation

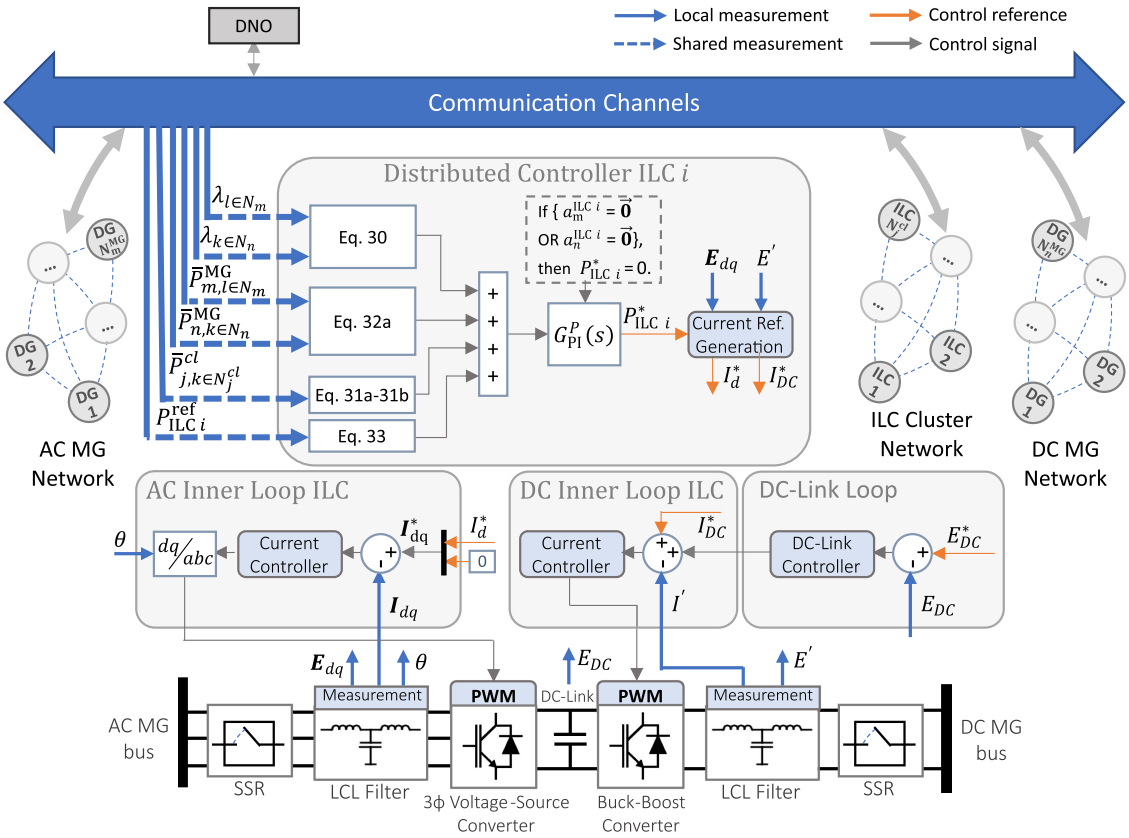

FIGURE 4. Schematic with an example of the proposed controller for the *i*th ILC in an ac/dc multi-MG.

depends on the maximum power of the ILC, so values $0 \rightarrow P_i^{\max}$ are possible. The parameter c_C can be chosen $0 \rightarrow 10c_L$ to be decoupled from the other control loops.

The weights for the tradeoff regularization can be designed as $\{h_m, h_n\} \in (0, 1]$. Also, as proposed in [11], the weights can be functions that emulate activation only when needed. Such function may take the form

$$h(\overline{P}_x) = k_h^1 \, e^{(k_h^2 \overline{P}_x)} \tag{34}$$

where the subindex x represents either interconnected MG (m or n), and k_h^1 and k_h^2 are defined parameters. The function starts with a small value $k_h^1 \approx 0$ that exponentially increments to 1 when the saturation of an MG reaches a critical level. The speed of this activation is decided with k_h^2. The input $\overline{P}_x$ might be bounded to avoid any implementation issues. Other alternatives of activation function, such as sigmoid or hyperbolic variations, are possible.

Regarding the finite-time convergence, the fractional exponents ν_1, ν_2, μ_1, μ_2, α_L, α_C, and α_P can start with 1 reducing their value until 0.5. As mentioned in Remark 6, values of 1 represent the removal of finite-time characteristics.

III. DESIGN OF DISTRIBUTED CONTROL FOR A MESHED MULTI-MG WITH BALANCING OF CLUSTERS OF ILCS

Occasionally, grid operators might need to induce some reductions in the total power transferred by the clusters of ICs to fulfill specific safety conditions or reduce the system's power losses. As demonstrated in [14], hard power constraints can be present in ILCs that perform global economic dispatch and ILC power sharing. Thus, approaches to safeguard the subgrids can be formulated by setting power references in selected ILCs. However, this is a complex process susceptible to jeopardizing the system's stability. To overcome this, a different approach can be taken by inducing a balancing between a group of ILC clusters; this avoids the necessity of power references that may lead the system to unnecessary saturations and instabilities.

From the design of the previous section, a novel ILC cluster balancing objective can be included to avoid ILC cluster saturations by adding a new control goal. Then, this section studies the feasibility of the implementation of such ILC cluster power balancing control and its effect on the overall system performance. For its formulation, it is applied the same principle shown in [11] for the MG saturation, i.e., a formulation relying on an average power observer.

A. FORMULATION OF ECONOMIC DISPATCH IN A MESHED AC/DC MULTI-MG WITH ILC POWER SHARING, HARD POWER CONSTRAINTS, SUBGRID POWER SATURATION, AND ILC CLUSTER POWER SATURATION

Similarly to the case of balancing the MG's average power to reduce the MG saturations described in [11], a control goal can be added to the objective function of 9 to perform the balancing of ILC cluster powers. Then, the objective function

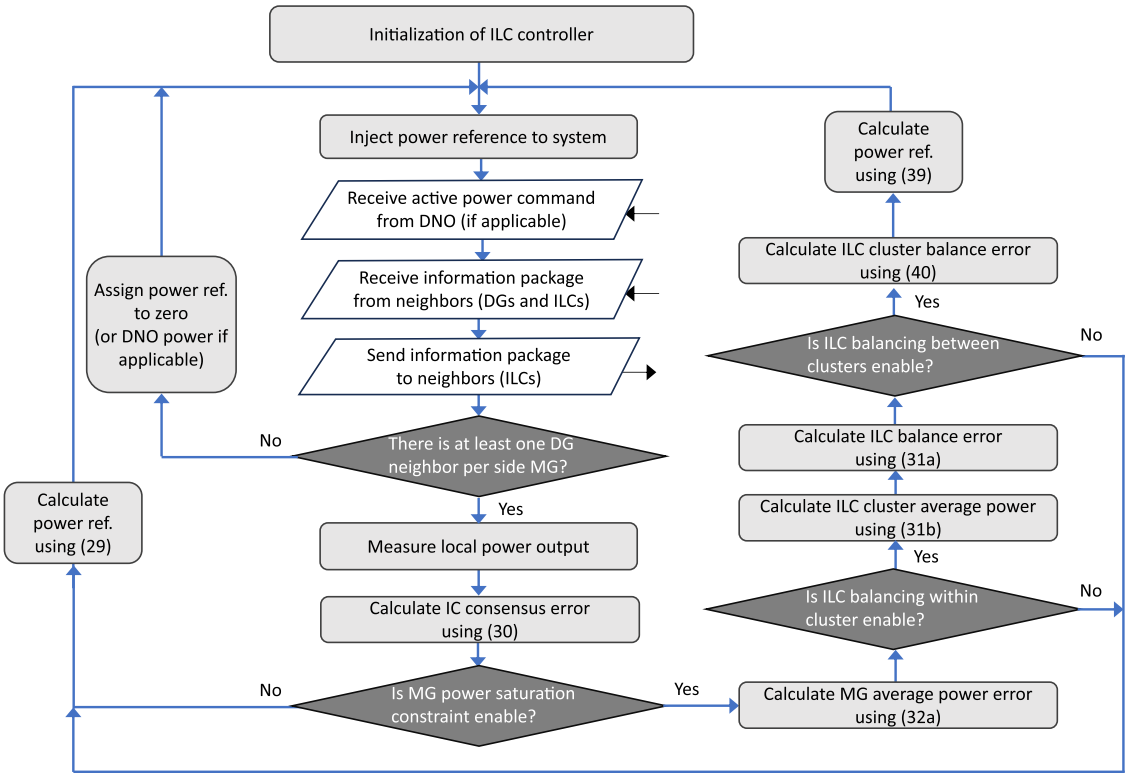

FIGURE 5. **Flowchart of the proposed multiobjective control strategy for ILCs in a meshed multi-MG system.**

can be expressed as

$$J = f_1(\boldsymbol{P}) + hf_2(\boldsymbol{P}) + wf_3(\boldsymbol{P}) + \Gamma f_4(\boldsymbol{P}) \tag{35}$$

where

$$f_4(\boldsymbol{P}) = \sum_{\substack{j=1\\k=1\\k\neq j}}^{N_{\text{cl}}} \left(d_j \overline{P}_j^{\text{cl}}(\boldsymbol{P}) - d_k \overline{P}_k^{\text{cl}}(\boldsymbol{P}) \right) \tag{36}$$

Γ is a weighting parameter to regulate the control goal of ILC cluster balancing, and $d_j \,\forall\, j \in \mathcal{N}_{\text{cl}}$ is a coefficient that determines the prioritization in the management of saturation/reserves of ILC clusters.

Thus, the stationary optimality condition gives

$$\lambda = \frac{\partial C_i(P_i)}{\partial P_i} + hg_1(\boldsymbol{P}) + wg_2(\boldsymbol{P}) + \Gamma g_3(\boldsymbol{P}) + \sigma_i^+ - \sigma_i^- + \sum_{j=1}^{N_{\text{ILC}}} (\Lambda_j^+ - \Lambda_j^-) \frac{\partial P_{\text{ILC}j}(\boldsymbol{P})}{\partial P_i} + \sum_{k=1}^{N_k} z_k \frac{\partial P_{\text{ILC}k}(\boldsymbol{P})}{\partial P_i} \tag{37}$$

where $g_3(\boldsymbol{P})$ is a linear function given by

$$g_3(\boldsymbol{P}) = \frac{\partial f_4(\boldsymbol{P})}{\partial P_i} \approx \sum_{\substack{j=1\\k=1\\k\neq j}}^{N_{\text{cl}}} \left(\frac{d_j}{N_j^{\text{cl}}} \sum_{l=1}^{N_j^{\text{cl}}} \frac{1}{P_{\text{ILC}l}^{\max}} \frac{\partial P_{\text{ILC}l}(\boldsymbol{P})}{\partial P_i} - \frac{d_k}{N_k^{\text{cl}}} \sum_{l=1}^{N_k^{\text{cl}}} \frac{1}{P_{\text{ILC}l}^{\max}} \frac{\partial P_{\text{ILC}l}(\boldsymbol{P})}{\partial P_i} \right). \tag{38}$$

B. DISTRIBUTED MULTIOBJECTIVE CONTROL FOR ILCS WITH ILC CLUSTER BALANCING USING CONSENSUS PROTOCOL

In this case, the communication between clusters of ILCs is assumed so that $\boldsymbol{a}_{\text{cl},k}^{\text{cl},j} \neq 0$ for all $k, j \in \mathcal{N}_{\text{cl}}$. Then, based on (29), the distributed controller for the ith ILC of the jth ILC cluster is

$$P_{\text{ILC}i}^* = (u_{\text{L}i} + u_{\text{C}i} + u_{\text{P}i} + u_{\Gamma i} + u_{\text{R}i}) G_{\text{PI}}^P(s) \tag{39}$$

with

$$u_{\Gamma i} = c_\Gamma \frac{\overline{P}_{j,i}^{\text{cl}}}{||\overline{P}_{j,i}^{\text{cl}}||} \sum_{l=1}^{N_{\text{cl}}} \sum_{k=1}^{N_l^{\text{cl}}} a_{\text{ILC}k}^{\text{ILC}i} \left(\Gamma_l \|\overline{P}_{l,k}^{\text{cl}}\| - \Gamma_i \|\overline{P}_{j,i}^{\text{cl}}\| \right) \tag{40}$$

where c_Γ is a convergence speed parameter, $\overline{P}_{l,k}^{\text{cl}}$ is the average power estimation of the lth cluster estimated by the kth ILC using (31b), and Γ_i is the tradeoff weight for the power sharing of ILC clusters regarding the IC consensus.

It can be seen from (40) that the ILC clusters' sign is managed. For the effects of the balancing, the subtraction is defined with only positive values. Then, the sign of the local ILC cluster is preserved and reincorporated at the end of the tracking error calculation.

A summary of the multiobjective control strategy for ILCs using (29)–(39) is given in Fig. 5. It describes the steps involving an arbitrary ILC of the multi-MG system.

TABLE 1. Comparison Summary of the Proposed Method Concerning Works in the Literature With Multiple MGs

Ref.	MG type	ILC type	Comm.	Control goal	Control technique	MG sat. const.	Cluster sat. const.	Comput. cost	ST oper. costs	LT oper. costs	Reliab.
[16], [19]	MM	none	centr.	ED	OPT	×	×	$ $ $	$	$ $ $	low
[17]	MM	none	centr.	ED	OPT	✓	×	$ $ $	$	$ $	low
[39]	SHM	SI	distr.	PS	ASC	×	×	$	$ $ $	$ $ $	high
[10], [13]	SHM	CI	distr.	PS	ASC	×	×	$	$ $ $	$ $ $	high
[40]	SHM	SI	mixed	ED	mixed	×	×	$ $	$ $	$ $ $	mid
[41]	SHM	SI	distr.	ED	ASC	×	×	$	$	$ $ $	high
[11]	SHM	SI	distr.	ED	FTC	✓	×	$	$ $	$	high
[14], [15], [42], [43]–[45]	SHM	CI	distr.	ED	ASC	×	×	$	$	$ $ $	high
[46]	SHM	CI	distr.	ED	ASC	×	✓	$ $	$	$ $	high
[47]	SHM	SI	distr.	ED	mixed	×	×	$ $	$	$ $ $	mid
[48]	SHM	CI	distr.	ED	mixed	×	×	$ $	$	$ $ $	mid
[20]	HMM	SI	centr.	ED	OPT	✓	×	$ $ $	$	$ $	low
[49]	HMM	CI	distr.	PS	ASC	×	×	$	$ $ $	$ $ $	mid
[50], [51]	HMM	SI	distr.	ED	ASC	×	×	$	$	$ $ $	high
This proposal	HMM	CI	distr.	ED	FTC	✓	✓	$	$ $	$	high

* MM: AC multi-MG, SHM: Single hybrid AC/DC MG, HMM: Hybrid AC/DC multi-MG, SI: Single ILC, CI: Cluster of ILCs, PS: Power-sharing, ED: Economic dispatch, OPT: Optimization, ASC: Asymptotic Consensus, FTC: Finite-Time Consensus, ST: Short-term, LT: Long-term.

C. GUIDELINE FOR THE SELECTION OF CONSENSUS PARAMETERS

Convergence parameter c_Γ can be chosen following the same logic than c_C, i.e., values $0 \rightarrow 10c_L$ such that the control actions u_{Li}, u_{Ci}, u_{Pi}, and $u_{\Gamma i}$ does not interfere between them.

Similarly to the h coefficients described in Section III, the Γ coefficients in (40) can be fixed or adjusted by an activation function, e.g., $\Gamma_i(\overline{P}_x^{\text{cl}}) = k_\Gamma^1 e^{k_\Gamma^2 \overline{P}_x^{\text{cl}}}$ where x is an ILC cluster; this ensures that a "penalty" compensation smoothly activates as the maximum power of an ILC cluster is reaching.

D. COMPARISON OF THE PROPOSED METHOD WITH THE LITERATURE

A comparison of the proposed control scheme against other approaches of the literature is presented in Table 1. Among the features used for the comparison are the communication topology, use of the constraints of the maximum power capacity of MGs and clusters of ILCs, computation cost, and operational costs. Regarding the costs, short-term and long-term cost components are displayed to clarify the benefits and drawbacks of each solution.

The reason for choosing short-term and long-term costs is that the proposed strategy promotes the deviation of ICs under certain overload/saturation conditions; this leads to higher operational costs in the short term (e.g., during failures) but reduces replacement and unscheduled shutdown costs. Then, the proposed multiobjective control safeguards critical infrastructures of the grid, such as converters, transformers, and distribution lines, which are prone to accelerate their aging or even fail due to overloading [26], [52], [53].

Optimization approaches could include penalty terms in the objective function, which could be analogous to the proposed solution. However, works that define penalties specifically for the saturation of MGs (as a whole) are scarce. Also, it is worth noting a point of contrast with optimizer methods: the computation time and effort required. The proposed distributed solution with adjustable weighting solves the economic dispatch problem seamlessly in the time/scale of secondary control whereas optimizers rely on iterative algorithms, such as particle swarm optimization or genetic algorithm, to find the optimal after several times the secondary control timescale.

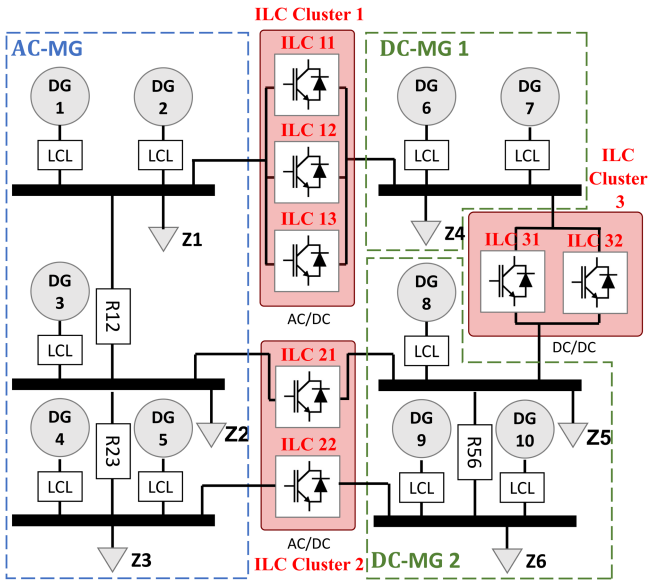

FIGURE 6. Simulated meshed multi-MG system.

IV. CASE STUDIES

To analyze the performance of the proposed terms for ILC controllers, simulations are carried out on the software PLECS [54]. For the simulations, an ac MG interconnected with two dc MGs is used; the system is depicted in Fig 6, and it is composed of five ac DGs, five dc DGs, three ac loads, and three dc loads. Also, there are three ILC clusters in the system, so the power flow of a meshed system can be studied.

TABLE 2. Economic Function Parameters in AC MG

Param.	DG1	DG2	DG3	DG4	DG5
a_{ci} [\$/kW^2]	0.264	0.444	0.400	0.500	0.250
b_{ci} [\$/kW]	0.067	0.111	0.100	0.125	0.063
c_{ci} [\$]	51.00	31.00	78.00	42.00	51.00

Note that ILCs of ILC Cluster #2 are located at different buses, generalizing the concept of ILC cluster.

The rating magnitude of voltage for the ac MG is 220 [V] per phase, whereas the rating frequency is 50 [Hz]. The rating voltage levels for the dc MG #1 and dc MG #2 are 400 [V] and 370 [V], respectively. The power ratings for the DGs are the following: $P_i^{\max} = 10$ [kW] $\forall i \in \mathcal{N}_{\text{AC}}$, $Q_i^{\max} = 3.33$ [kVAR] $\forall i \in \mathcal{N}_{\text{AC}}$, and $P_i^{\max} = 10$ [kW] $\forall i \in \mathcal{N}_{\text{DC}}$. The power capacities of ILCs are: $P_{\text{ILC11}}^{\max} = 11, 0$[kW], $P_{\text{ILC12}}^{\max} = 7, 7$[kW], $P_{\text{ILC13}}^{\max} = 14, 3$[kW], $P_{\text{ILC21}}^{\max} = 10, 0$[kW], $P_{\text{ILC22}}^{\max} = 7, 0$[kW], $P_{\text{ILC31}}^{\max} = 9, 0$[kW], and $P_{\text{ILC32}}^{\max} = 6, 3$[kW]. The communication matrices and vectors are

$$A^{\text{AC}} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}, \; A_{\text{AC}}^{\text{cl},1} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A^{\text{DC1}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \; A_{\text{DC1}}^{\text{cl},1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \; A_{\text{DC2}}^{\text{cl},1} = 0_{3\times 3}$$

$$A^{\text{DC2}} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \; A_{\text{AC}}^{\text{cl},2} = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{\text{DC1}}^{\text{cl},2} = 0_{2\times 2}, \; A_{\text{DC2}}^{\text{cl},2} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A_{\text{AC}}^{\text{cl},3} = 0_{2\times 5}, \; A_{\text{DC1}}^{\text{cl},3} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \; A_{\text{DC2}}^{\text{cl},3} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$A_{\text{cl},1}^{\text{cl},1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \; A_{\text{cl},2}^{\text{cl},1} = 0_{3\times 2}, \; A_{\text{cl},3}^{\text{cl},1} = 0_{3\times 2}$$

$$A_{\text{cl},1}^{\text{cl},2} = (A_{\text{cl},2}^{\text{cl},1})^{\text{T}}, \; A_{\text{cl},2}^{\text{cl},2} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \; A_{\text{cl},3}^{\text{cl},2} = 0_{2\times 2}$$

$$A_{\text{cl},1}^{\text{cl},3} = (A_{\text{cl},3}^{\text{cl},1})^{\text{T}}, \; A_{\text{cl},2}^{\text{cl},3} = (A_{\text{cl},3}^{\text{cl},2})^{\text{T}}, \; A_{\text{cl},3}^{\text{cl},3} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The economic function parameters of the DGs in the ac MG are described in Table 2. For the DGs in the dc MGs, the values are 3/4 of the ones used in the ac MG. The ILC control parameters are $k_p^P = 0.25$, $k_i^P = 2.35$, $c_{\text{L}} = 1000$, $c_{\text{C}} = 2000$, $c'_{\text{C}} = 5$, $c_{\text{P}} = 0.1 P_{\text{ILC}i}^{\max}$, $c_\Gamma = 3000$, $k_h^1 = 10^{-7}$, and $k_h^2 = 16$. The finite-time gains are $\alpha_{\text{L}} = 0.8$, $\alpha_{\text{C}} = 1.0$, and $\alpha_{\text{P}} = 1.0$. For briefness in the analysis of simulation curves, power references to ILCs are omitted, i.e., $c_{Ri} = 0$; the interested reader is conducted to [14] to see the behavior of the economic dispatch and ILC power sharing with this condition. The remaining parameters of the system are shown in Table 3.

TABLE 3. System Parameters of multi-MG

Parameter	Value	Parameter	Value
Load Z1	15 [Ω]	Line R1-2	1 [mΩ]
Load Z2	33 [Ω]	Line L1-2	2.50 [mH]
Load Z3	12 [Ω]	Line R2-3	1 [mΩ]
Load Z4	11 [Ω]	Line L2-3	2.50 [mH]
Load Z5	8 [Ω]	Line R5-6	0.50 [Ω]
Load Z6	16 [Ω]	Line L5-6	2.50 [mH]

For the performance analysis under challenging operation, i.e., load impacts, saturations, and time delays, four sets of cases of study are used to test each control goal. The cases of study are defined as follows.

A. CASE 1.A. ECONOMIC DISPATCH WITH LOAD IMPACTS

In this test, the ILCs apply the controller of (30) for IC synchronization. The subgrids are subdued to load impacts to analyze the transient and steady-state performance: at $t = 23$ [s], Z3 changes from 7.69 to 4.69[Ω], then, at $t = 43$ [s], Z6 changes from 12.63 to 8.47[Ω], finally, at $t = 63$ [s], Z6 changes again to 12.63 from 8.47[Ω].

B. CASE 1.B. ECONOMIC DISPATCH WITH LOAD IMPACTS AND ILC CLUSTER SATURATION

In this case, the maximum power capacity of the ILC Cluster #3 is reduced to produce saturation under certain load impacts. The ILCs of Cluster #3 have $P_{\text{ILC31}}^{\max} = 1800$[W] and $P_{\text{ILC32}}^{\max} = 415, 8$[W].

C. CASE 1.C. ECONOMIC DISPATCH WITH LOAD IMPACTS AND COMMUNICATION LOSSES

In this test, the economic dispatch is tested when one ILC cluster stops its power transfer. The system starts with the same conditions as Case 1.b., and then, the communication between the side MGs and the ILCs in Cluster #3 is interrupted. Under this condition, only two ILC clusters are operative for the economic dispatch. Sufficient power capacity is available in the remaining ILC clusters.

D. CASE 1.D. ECONOMIC DISPATCH WITH LOAD IMPACTS AND COMMUNICATION TIME DELAYS

In this test, different time delays are incorporated into the communication links between DGs and ILCs. For briefness, only homogeneous delays are incorporated for all communication links. Delays of 0 [s], 0.125 [s], and 0.25 [s] are analyzed using the system of Case 1.b.

E. CASE 2. ECONOMIC DISPATCH WITH MG SATURATION AND LOAD IMPACTS

The ILCs perform the control actions in (30) and (32). For the simulation, the maximum power capacity of some ILCs and DGs is reduced to produce the saturation of one MG. Specifically, $P_{\text{ILC31}}^{\max} = 1800$[W], $P_{\text{ILC32}}^{\max} = 415, 8$[W], $P_6^{\max} = 4400$[W], $P_7^{\max} = 4400$[W], and $P_8^{\max} = 4000$[W]. Values of $h_m = 0.2$ are used for ILCs in Cluster #1, as well as the adjustable expression $h_m(\overline{P})$ of [11] but multiplied by 0.1. The parameters of (34) are selected such that $h(1.0) = 0.9$, i.e., nearly 90% of the ILC capacity when an MG is at maximum capacity. The same load impacts as previous simulations are used. A time delay of 0.125 [s] is also used.

F. CASE 3.A. ECONOMIC DISPATCH WITH ILC POWER SHARING AND LOAD IMPACTS

This test implements the ILC power balancing inside ILC clusters. It uses the same setup as Case 1.b., i.e., without MG saturations but with the new term in (31) in the ILCs' controller. The test looks to validate the simultaneous synchronization of global IC and the ILC powers inside a cluster.

G. CASE 3.B. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND LOAD IMPACTS

In this test, the term (31) is included in the ILCs' controller with saturations. The same setup as Case 2 with the adjustable $h_m(\overline{P})$ is used, i.e., there is saturation in dc MG #1 and ILC Cluster #3 with a time delay between DGs and ILCs.

H. CASE 3.C. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND COMMUNICATION LOSSES

This test performs Case 3.b. but with a communication loss between the ILC #11 and the ILC #12.

I. CASE 3.D. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND COMMUNICATION TIME DELAYS

This test performs Case 3.a. but with a homogeneous time delay in all ILCs. Delays of 0, 0.125, and 0.25 [s] are used. The same antiwindup with reset scheme of [11] is implemented to deal with steady-state errors. Also, a subcase is tested where the coupling gains of Cluster #3 are reduced by half to evaluate the decoupling with other control loops.

J. CASE 4.A. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND ILC SATURATION CONSTRAINT

This test uses (39) in the ILCs. The communication vectors between clusters of ILCs are nonzero, taking the values:

$$A_{cl,2}^{cl,1} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_{cl,3}^{cl,1} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \text{and} \quad A_{cl,3}^{cl,2} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The same MG conditions of Case 3.b. are used with a delay of 0.125 [s] included in the ILC power-sharing communications. Also, Γ_j values of 0, 0.5 and the adjustable $\Gamma_j(\overline{P}_{\text{ILC}})$ are used simultaneously for all the ILC clusters.

K. CASE 4.B ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, ILC SATURATION CONSTRAINT, AND COMMUNICATION LOSSES

In this test, communication losses are included in the communication between Cluster #1 and Cluster #2. To represent the worst-case scenario, all the communication links are disconnected between these two clusters, i.e., $A_{cl,2}^{cl,1} = 0_{3\times 2}$. The rest of the system parameters are the same as Case 4.a. with the adjustable $\Gamma_j(\overline{P}_{\text{ILC}})$.

L. CASE 4.C ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, ILC SATURATION CONSTRAINT, AND COMMUNICATION TIME DELAYS

In this test, delays are included in the cluster communication, which are 0, 0.125, and 0.25 [s]. The simulation setup is the same as in Case 4.a. with the adjustable $\Gamma_j(\overline{P}_{\text{ILC}})$.

V. RESULTS AND DISCUSSION

A. CASE 1.A. ECONOMIC DISPATCH WITH LOAD IMPACTS

Fig. 7 summarizes the results for the application of (30) in the ILCs of Fig. 6. The system starts with droop control in the DGs. Secondary control is activated at $t = 0.3$ [s] for voltage restoration and IC consensus, whereas the secondary control for frequency restoration is activated at $t = 1.8$ [s].

It can be noted from Fig. 7(a) that all the individual ICs of DGs in the multi-MG synchronize to the same value when the ILC control is enabled, at $t = 3$ [s], even though there are no direct communications between DGs of different MGs or between ILCs. Also, seamless transitions of ICs during the load impacts can be observed.

As for the ILCs' powers, Fig. 7(b) shows a power distribution that depends on DGs' location, communication links with ILCs, and cost functions. Similarly, Fig. 7(c) shows how the combined power of DGs varies over time as a consequence of the economic dispatch. It can be seen that the dc MG #2 has the highest overshoot on average, shown by its load impacts occurring in $t = 43$ [s] and $t = 63$ [s].

MGs' average voltages are shown in Fig. 7(d) and (e), whereas the frequency of the only ac MG is shown in Fig. 7(f). It can be seen that the secondary control in each MG is performed seamlessly (each local average voltage reaches its voltage reference), and it does not modify, nor does it get modified by the ILC control loop during load impacts. A similar situation is observed for the frequency, where the locally measured frequency stabilizes to the reference every time there is a load impact.

B. CASE 1.B. ECONOMIC DISPATCH WITH LOAD IMPACTS AND ILC CLUSTER SATURATION

The results for saturation of ILC Cluster #3 are shown in Fig. 8. Notably, Fig. 8(a) illustrates, once again, how the DGs and ILCs manage to equalize the ICs. There are no noticeable

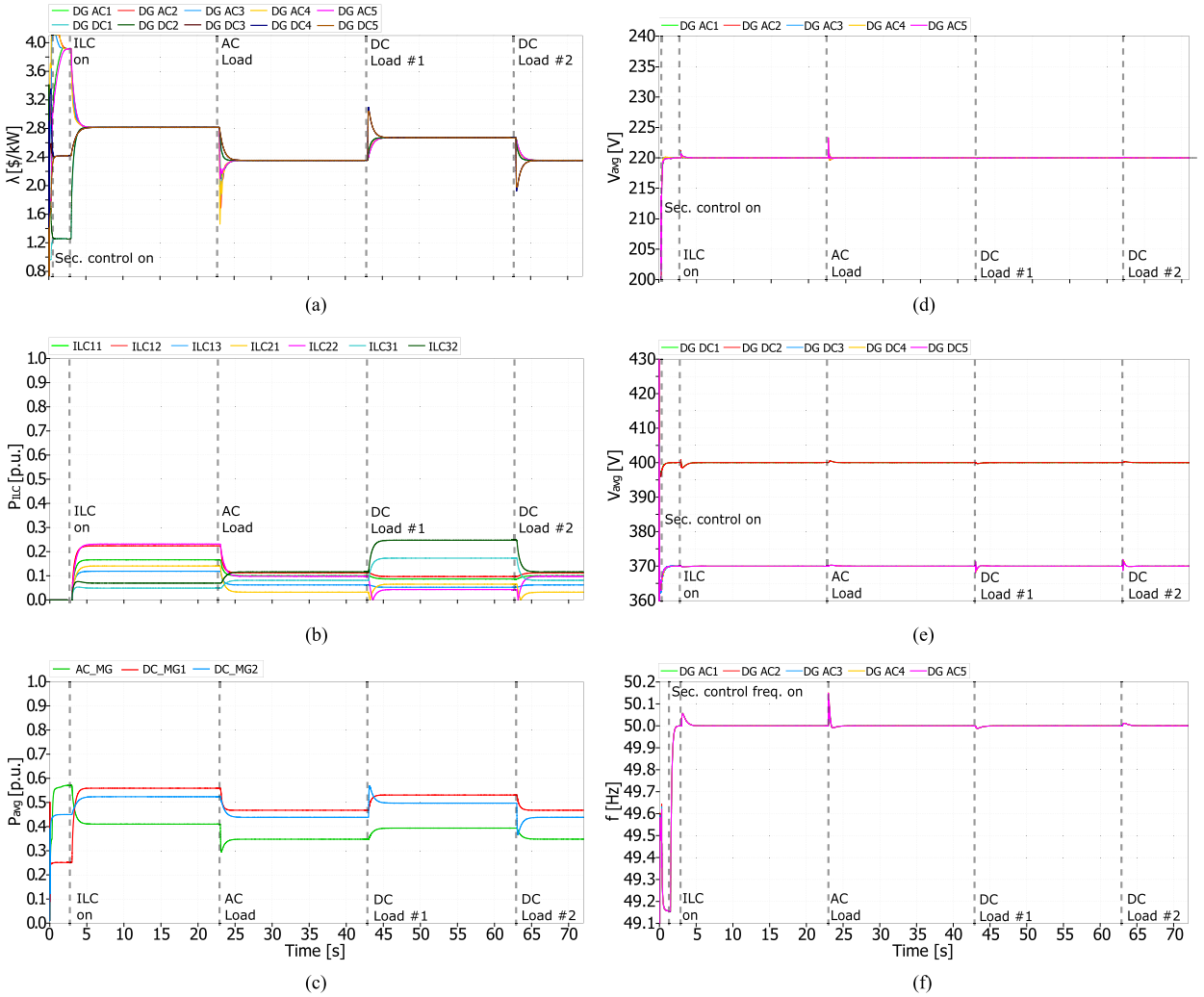

FIGURE 7. Simulation results for Case 1.a. (a) IC of DGs in multi-MG. (b) Absolute power of ILCs. (c) Average power of MGs. (d) Average voltage of AC MG. (e) Average voltage of dc MG #1 and dc MG #2. (f) Frequency of AC MG.

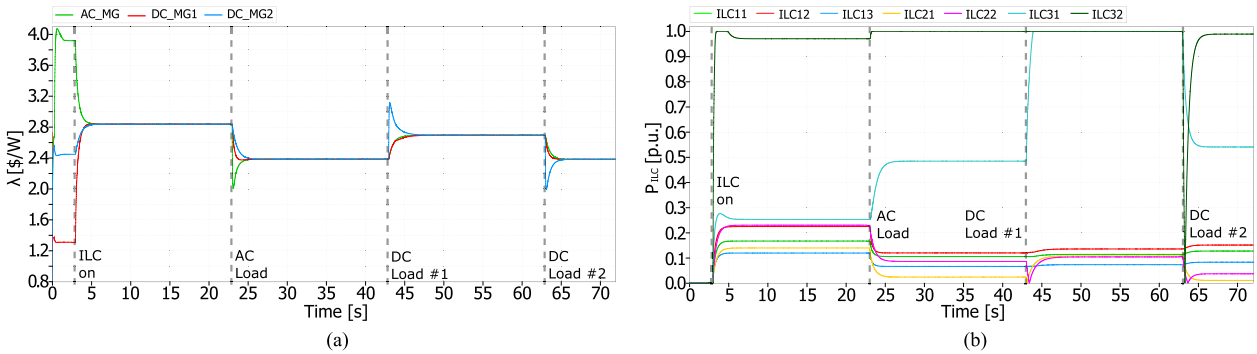

FIGURE 8. Simulation results for Case 1.b. (a) Average IC of MGs. (b) Absolute power of ILCs.

differences in the transients and steady-state values compared with Fig. 7(a). It should be pointed out that the curves in Fig. 8(a) are average IC values, i.e., all the ICs of DGs in each MG are used for an average calculation before plotting; this means that the overshoots in Fig. 8(a) look lower than Fig. 7(a).

As shown in Fig. 8(b), since the beginning of the ILC control action, at $t = 3$ [s], ILC #32 almost saturates. After the first load impact on the ac side ($t = 23$ [s]), ILC #32 saturates and ILC #31 reaches roughly its mid capacity. The situation is more critical after the first load impact in dc MG #2, where both ILC #31 and ILC #32 saturate. In this case, the economic

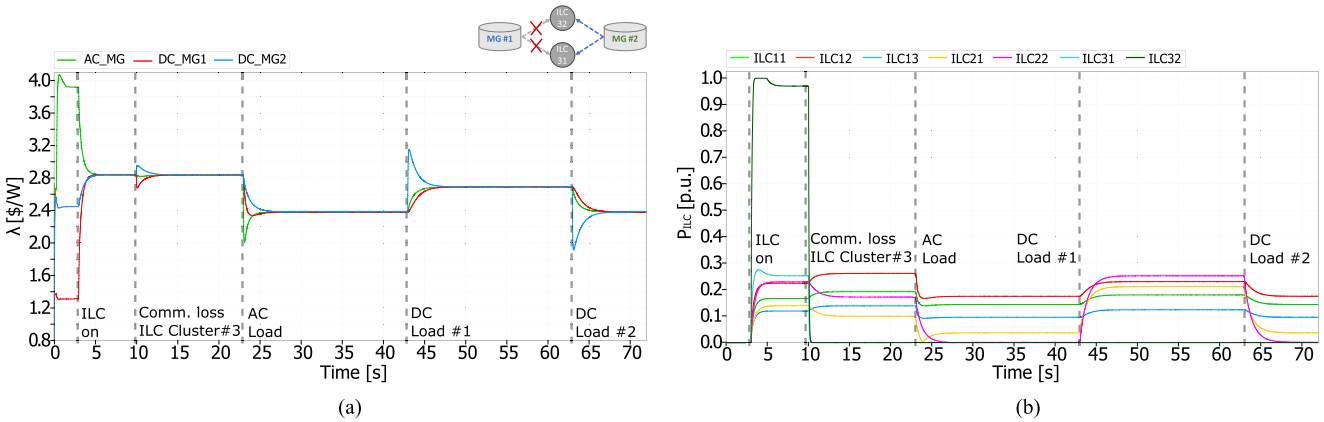

FIGURE 9. Simulation results for Case 1.c. (a) Average IC of MGs. (b) Absolute power of ILCs.

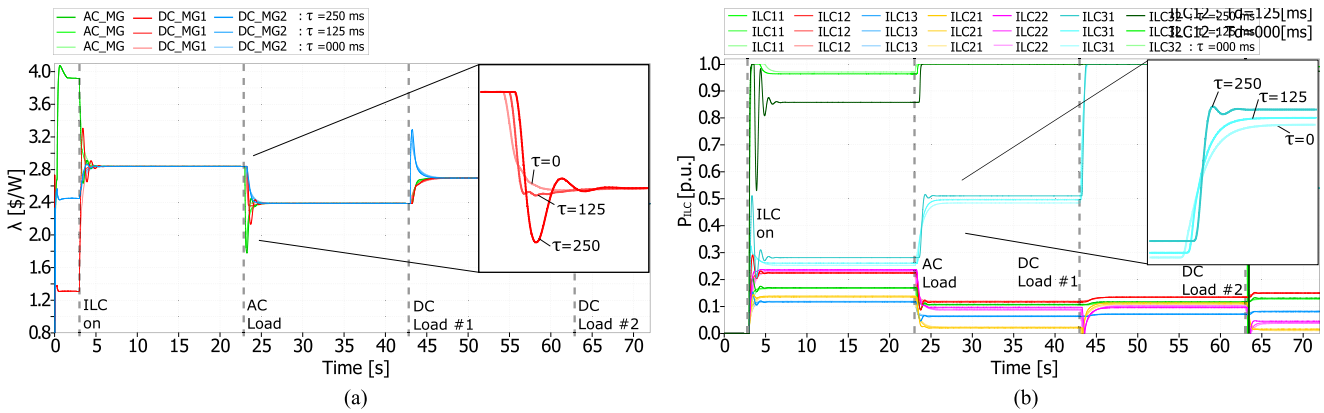

FIGURE 10. Simulation results for Case 1.d. (a) Average IC of MGs. (b) Absolute power of ILCs.

dispatch is still possible due to the meshed multi-MG topology, which has another path to deliver the power transfers. It should be noted that if there is insufficient capacity in any other ILC cluster, there will be desynchronization in the ICs.

C. CASE 1.C. ECONOMIC DISPATCH WITH LOAD IMPACTS AND COMMUNICATION LOSSES

The results of this test are presented in Fig. 9. In Fig. 9(a), the ILC controllers can keep the consensus of ICs after the communication loss in the ILC Cluster #3 at $t = 10$ [s], reaching the same steady-state values as Cases 1.a and 1.b. However, the frequency bandwidth of the control is reduced as it has an increased settling time and overshoot. This is mainly because there is one less path for the power to flow, so it can be seen as a virtual load impact on the rest of the ILCs.

In Fig. 9(b), after $t = 10$ [s], all the ILCs achieve new operation points compared with Case 1.b. It is worth remembering that ILC #31 and ILC #32 clamped to zero their power output after $t = 10$ [s] as it is programmed to happen when there is no communication from one MG side nor a DNO power command (see schematic of Fig. 5). A slower transient is observed, mainly as a consequence of the IC dynamics changes, as described in previous paragraphs.

D. CASE 1.D. ECONOMIC DISPATCH WITH LOAD IMPACTS AND COMMUNICATION TIME DELAYS

Communication delays are analyzed in Fig. 10 for the controller in (30). In Fig. 10(a), it is shown how time delays slowly increased the ILCs' power overshoot, although the variations are very low. Similar trends are observed in Fig. 10(b), where the delays also induce small steady-state variations in the absolute power of ILCs. The most significant case is ILC #32, especially in $3 < t < 23$ [s], where the difference caused by the 250 [ms] time delay is above 10% concerning the case without delays. For the rest of the ILCs, time delays provoke steady-state deviations of less than 1% .

E. CASE 2. ECONOMIC DISPATCH WITH MG SATURATION AND LOAD IMPACTS

The main results for the multiobjective ILC controller with MG saturations are summarized in Fig. 11. Overall, due to the reduction in the converters' power capacities of DGs, all charts reflect an increase in the overshoot during transient states, i.e., after the activation of the ILCs and load impacts, compared with the results of Case 1.b.

In Fig. 11(a), it can be seen that after the activation of the saturation constraint, at $t = 13$ [s], a higher value for the h_m coefficient produces an increase in the transient oscillations

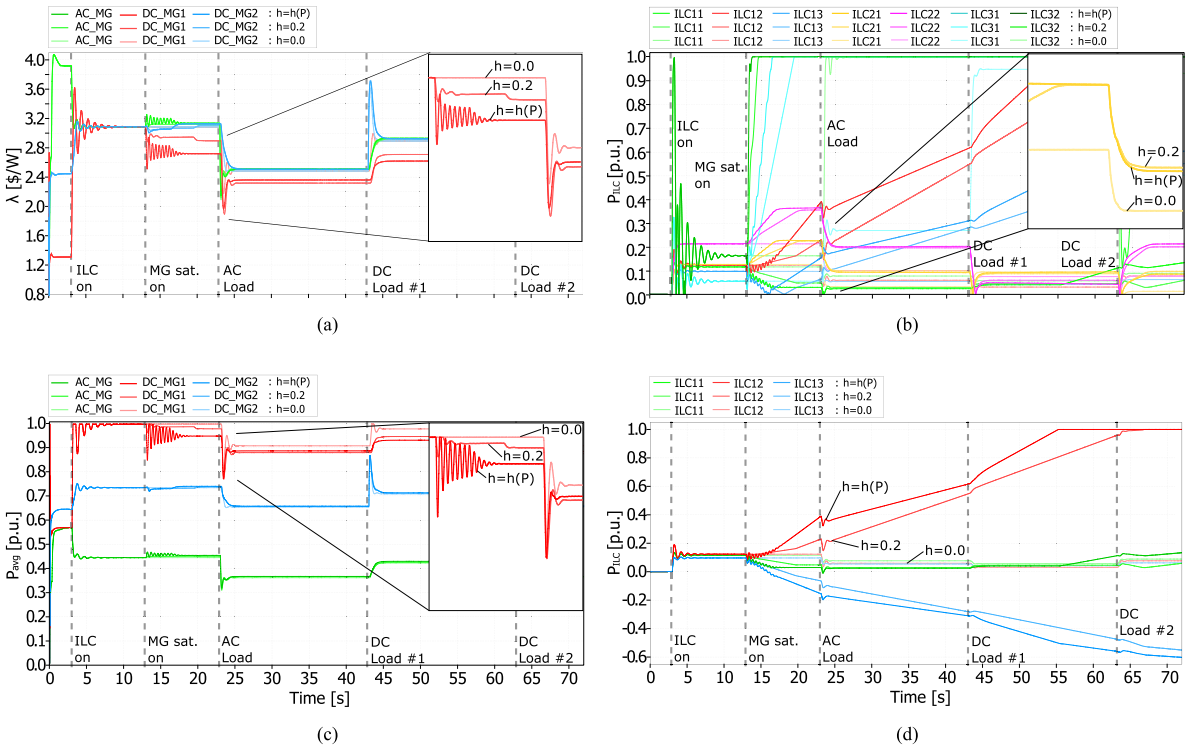

FIGURE 11. Simulation results for Case 2. (a) Average IC of MGs. (b) Absolute power of ILCs. (c) Average power of MGs. (d) Power of ILCs in Cluster #1.

of ICs. This phenomenon vanishes after a transient of around 5 [s]. As for the steady-state variations, the ICs show different values compared with Case 1, i.e., there is a desynchronization of ICs. Similarities regarding the sensibility to h_m and transient state oscillations are observed in Fig. 11(c).

In Fig. 11(b), an increasing behavior of the absolute powers of ILC #12 and ILC #13 can be seen after $t = 13$ [s]. Indeed, in Fig. 11(d), it can be seen that these powers neutralize each other (with different signs), so an equilibrium is reached in the total power transferred by the ILC cluster. However, the power of both ILCs gets saturated, so this behavior is not ideal for real applications. In the steady state, Fig. 11(b) shows that the saturated MG (dc MG #1) get reductions in its average power, as expected. The adjustable h_m provides the greater reduction in this case; values of $h_m > 0.4$ could achieve the same reduction in the saturated MG but at the cost of deteriorating the IC synchronization continuously. The steady-state variations are low for ac MG #1 and dc MG #2, primarily because they do not use more than 80% of their capacity. For ILC Cluster #2 (yellow curves), which does not directly interconnect the saturated MG, the steady-state values are not saturated and the difference between $h = 0.2$ and $h = h(P)$ is insignificant.

The results presented in this case study suggest that the strategy, despite being able to reserve power from saturated MGs, has problems of divergence inside the ILCs of a cluster. This is ineffective since some of the ILCs operate saturated unnecessarily. Therefore, additional control actions should be taken inside the ILC clusters to allow the power constraint to be activated in real-world scenarios.

F. CASE 3.A. ECONOMIC DISPATCH WITH ILC POWER SHARING AND LOAD IMPACTS

Fig. 12 shows the performance of the ILC power balancing in the multi-MG. In Fig. 12(a), the ICs of the MGs in the system synchronized in the steady state. The transient-state behavior of ICs slightly changes compared with Fig. 8(a); the same overshoot and settling time are observed.

The absolute power of ILCs is represented in Fig. 12(b). It can be seen that the ILCs of each cluster equalize their per unit powers after $t = 5.5$ [s], where the ILC power sharing is activated. It is worth noting that, although there is saturation in Cluster #3, the other ILC clusters behave as expected, sharing the power between their ILCs. So, it is confirmed that the ILC power sharing could be executed with a similar bandwidth to the economic dispatch without affecting its performance.

G. CASE 3.B. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND LOAD IMPACTS

Fig. 13 shows the performance of the ILC power balancing in the multi-MG with the saturation of dc MG #1. From Fig. 13(a), the same performance can be seen for the average ICs compared with Fig. 11(a) with the adjustable h_m coefficient. In general, the ILC power sharing does not reflect changes in the IC consensus nor the net power flow between MGs. Since its application in $t = 3$ [s], there are no visible changes in this case of study.

Based on the response of Fig. 11(b) of Case 2, a different scenario is seen in Fig. 13(b). The ILC powers are now synchronized after the activation of the ILC power sharing, at

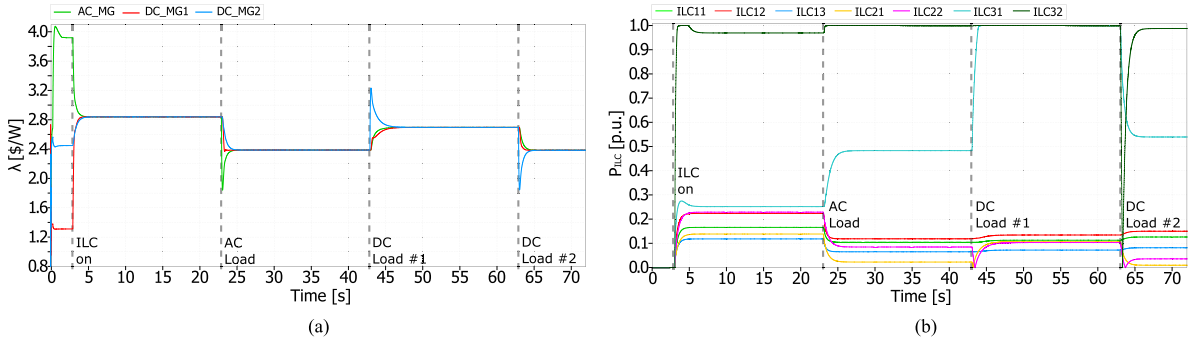

FIGURE 12. Simulation results for Case 3.a. (a) Average IC of MGs. (b) Absolute power of ILCs.

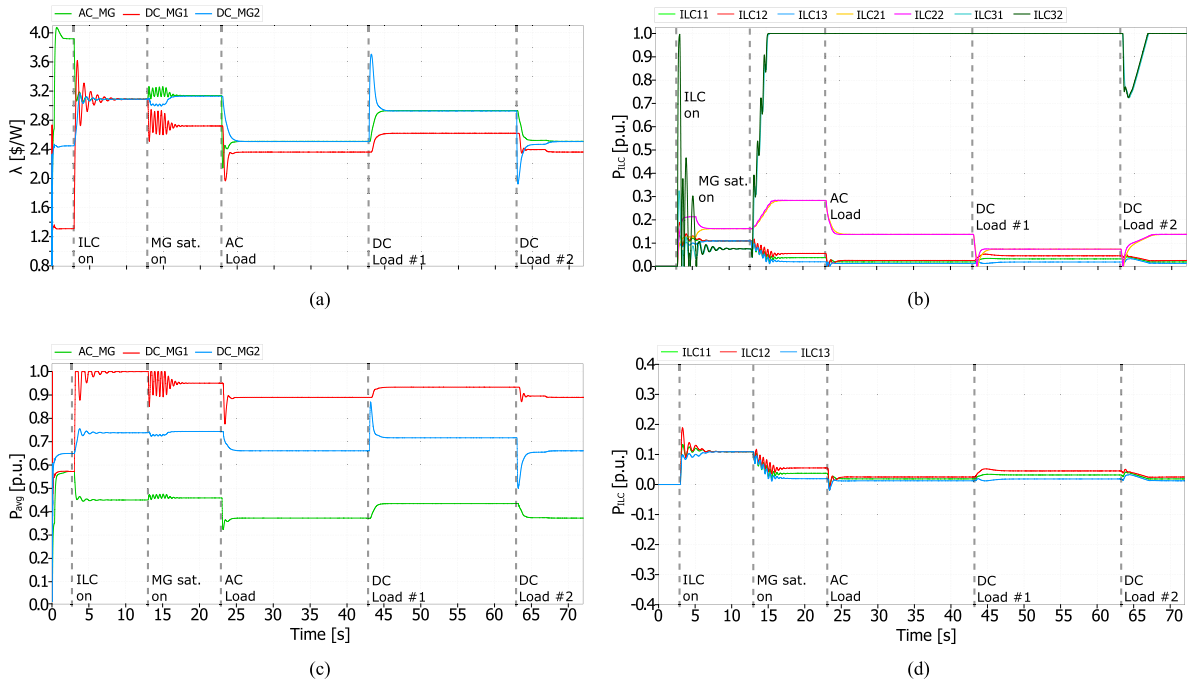

FIGURE 13. Simulation results for Case 3.b. (a) Average IC of MGs. (b) Absolute power of ILCs. (c) Average power of MGs. (d) Power of ILCs in Cluster #1.

$t = 3$ [s], and the consensus values for the ILC clusters are reached [instead of the steady-state values seen in Fig. 11(b)].

The case of the ILC Cluster #1 with the ILC power balancing is depicted in detail in Fig. 13(d). Here, the increasing behavior of ILC #12 and ILC #13 seen in Case 2 is nonexisting. Therefore, the ILC power sharing aids in solving the problem of saturation of ILCs when there is a saturation of one of the side MGs and the controller of (32) is activated.

H. CASE 3.C. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND COMMUNICATION LOSSES

The charts in Fig. 14 show the results of a communication failure in a communication link between ILCs. In Fig. 14(a) and (b), the same trends of Case 3.b. are visualized; negligible changes can be perceived in the transient states of the ILC powers, particularly, in the ILCs of Cluster #1 where the communication link failure occurred. It can be confirmed that as long as a spanning tree exists in the communication network, the consensus is reached.

I. CASE 3.D. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND COMMUNICATION TIME DELAYS

Simulation results using time delays in the ILC power sharing are shown in Fig. 15. Overall, Fig. 15(a) and (b) exposes a high level of transient and steady-state oscillations when the time delay rises close to 250 [ms]. This situation occurs due to the initial tuning that decouples ILC's IC consensus and ILC's power sharing; the ILC power-sharing loop was designed to be fast, however, this fact makes the ILC stability vulnerable to time delays (especially because of the use of a distributed observer).

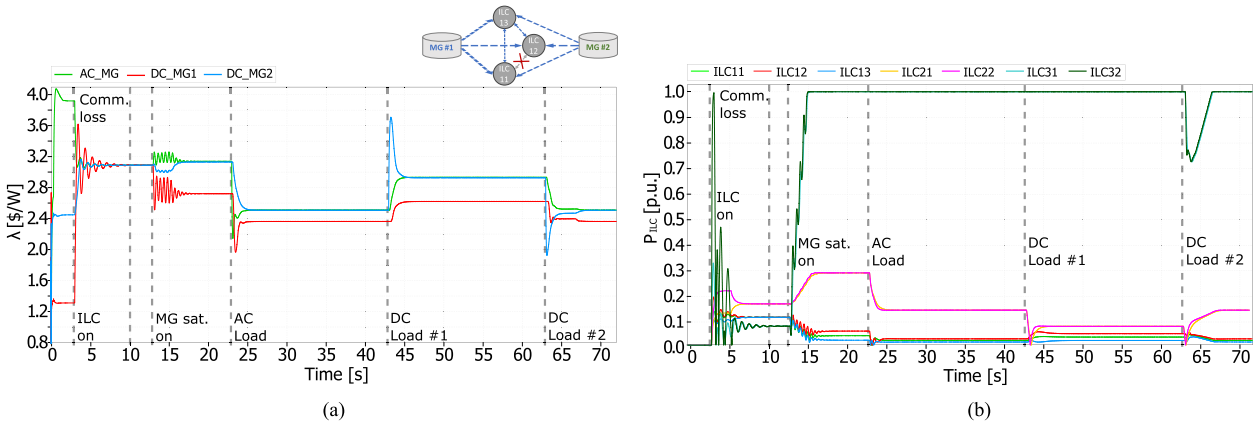

FIGURE 14. Simulation results for Case 3.c. (a) Average IC of MGs. (b) Absolute power of ILCs.

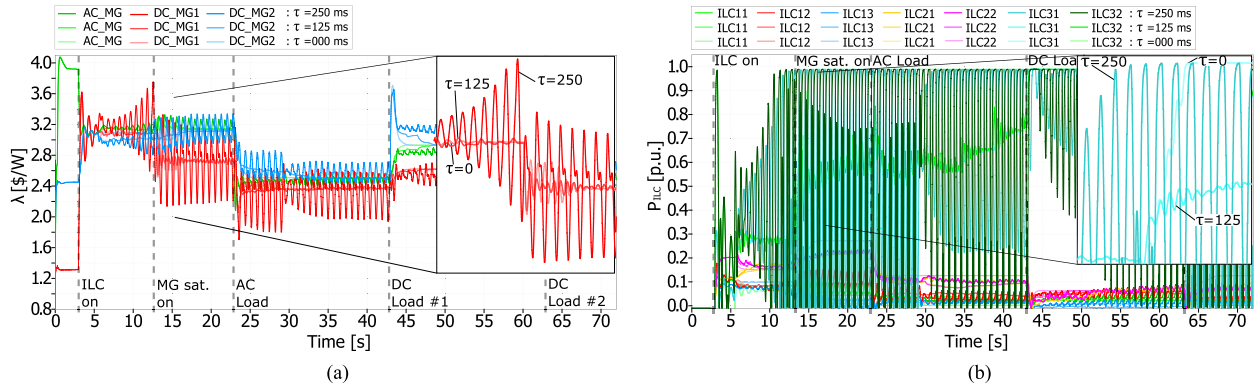

FIGURE 15. Simulation results for Case 3.d. (a) Average IC of MGs. (b) Absolute power of ILCs.

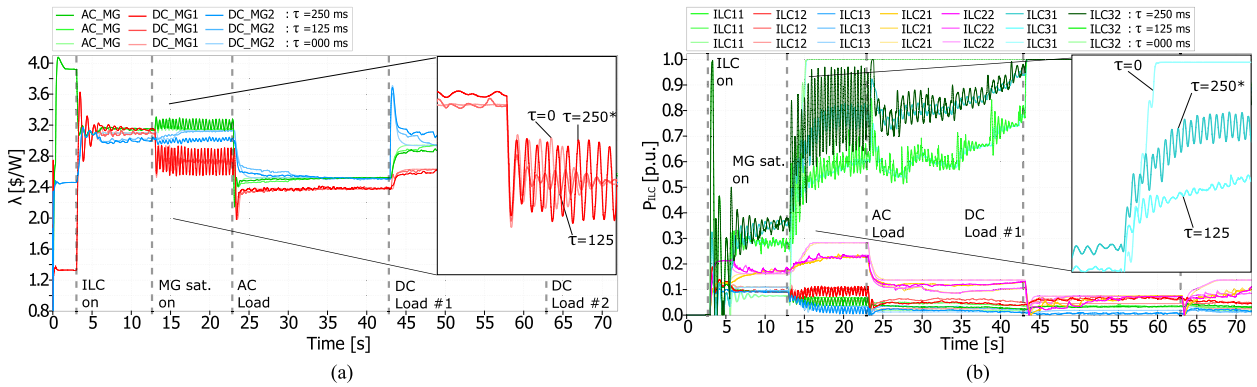

FIGURE 16. Simulation results for Case 3.d. when c_C is reduced by half in Cluster #3. (a) Average IC of MGs. (b) Absolute power of ILCs.

Fig. 15(a) shows initial differences in the IC consensus after each load impact; moreover, these differences are proportional to the time delay applied. Time delays applied to the distributed power observer of (31b) create steady-state errors, which are resolved once the antiwindup algorithm of [14] executes a reset. In general, these steady-state errors are subsequently reflected in the IC consensus. Notably, at the short transient of $t = 13$ [s], the case with 125 [ms] seems to have a better transient state once the MG saturation control is activated. This is because the distributed observer steady-state errors intervene and slow the control bandwidth of the IC consensus, producing damping.

Despite the consensus between mean steady-state values of IC is being reached on every occasion, there are persistent oscillations in the steady state for the delay of 250 [ms]. In the case with a delay of 125 [ms], the steady-state oscillations in IC are almost negligible, except for the dc MG #1. The absolute powers of ILCs, shown in Fig. 15(b), depicted with more detail the oscillatory tendency described in Fig. 15(a). An extreme oscillatory behavior is perceived in Cluster #3,

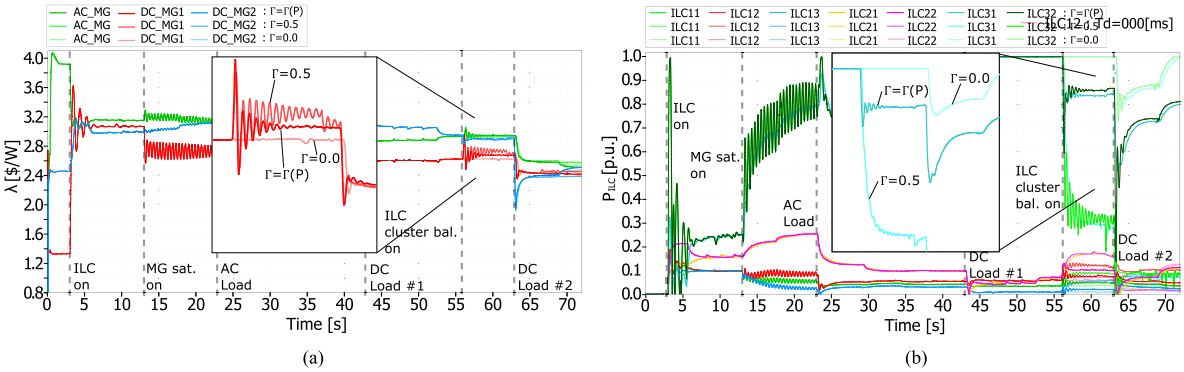

FIGURE 17. Simulation results for Case 4.a. (a) Average IC of MGs. (b) Absolute power of ILCs.

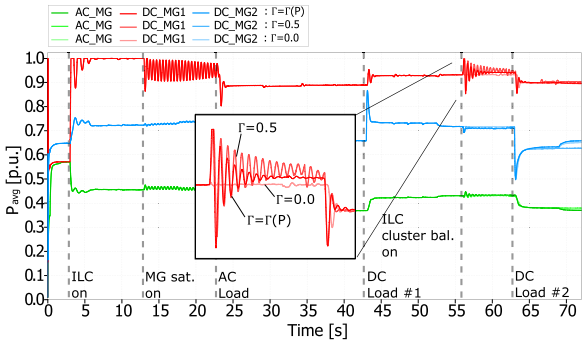

FIGURE 18. Simulation results for average power of MGs during Case 4.a.

where the time delay of 250 [ms] causes the biggest oscillation amplitude in ILC #32, followed by ILC #31. The case of the time delay of 125 [ms] presents a significant steady-state deviation compared to the base case of $\tau = 0$ [ms]. This steady-state deviation is proportional, giving roughly half of the steady-state deviation observed for the mean values of the ILC powers when compared with the case of $\tau = 250$ [ms].

Another set of charts with simulation results but with a reduction by half in the proportional gain c_C of ILC #31 and #32 is presented in Fig. 16(a) and (b). The new charts keep the curves for the $\tau = 0$ [ms] and $\tau = 125$ [ms] cases but update the case of $\tau = 250$ [ms] with the new proportional gains. Overall, a great oscillation can still be seen during $13 < t < 23$, where the MG saturation constraint interfered with the ILC economic dispatch. The reason for this persistency in steady-state oscillations for $\tau = 250$ [ms] is attributed to couplings between economic dispatch, MG saturation, and ILC power-sharing control actions. Despite the results shown, decoupling can be obtained by reducing the control bandwidth of the control loops, e.g., by reducing some of the coupling gains c_L, c_P, and c_C.

J. CASE 4.A. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, AND ILC SATURATION CONSTRAINT

Results for the ILC cluster balancing algorithm are shown in Fig. 17. Overall, oscillations can be seen when the balancing of ILC clusters is activated, at $t = 56$ [s]. However, such transient oscillations are damped quickly by the control strategy; they do not appear again even after load changes as the one in $t = 63$ [s]. On the one hand, the effects of the ILC cluster power sharing inside the ILC power sharing are negligible. Whatever condition of Γ, the ILCs of the same cluster did not deteriorate their power sharing. On the other hand, as can be seen in Fig. 18, ILC cluster balancing does have a minor effect on the total average power of the MGs, which inevitably deteriorates the IC synchronization but in a small amount. This is because both conditions, the saturation of dc MG #1 and the saturation of ILC Cluster #3 are present, so the controller adjusts the IC regulation according to the weight parameters.

The ICs are shown in Fig. 17(a). After the transient oscillations got stabilized (before $t = 63$ [s]), parameter ICs reached different steady-state values compared with Fig. 13(a) of Case 3.b., with differences near 10% depending on the Γ value. The dc MG #1 case presents the biggest oscillations, especially when $\Gamma = 0.5$. After the reset of the antiwindup at $t = 68$ [s], the curve of $\Gamma = \Gamma(P)$ obtained the closest ICs values to the condition without the activation of the ILC cluster consensus ($\Gamma = 0$).

The ILC powers are shown in Fig. 17(b). It can be seen that after the activation of the ILC cluster balancing, at $t = 56$ [s], the ILC powers get closer. In the steady state, the results show that, in general, the ILC clusters get closer to each other the more weight the Γ coefficient gets. The biggest differences are shown in ILC #31 and ILC #32, which are part of the saturated ILC cluster.

K. CASE 4.B. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, ILC SATURATION CONSTRAINT, AND COMMUNICATION LOSSES

The results for a complete loss of communications between the ILCs of the clusters #1 and #2 are shown in Fig. 19. The observed transient state and steady-state behaviors of the curves of average ICs [see Fig. 19(a)] and absolute power of ILCs [see Fig. 19(b)] present no changes concerning the results of Case 4.a. with $\Gamma = \Gamma(P)$. This behavior is expected since the communication system still has a spanning tree, i.e., the communication flows from ILC cluster #1 to ILC

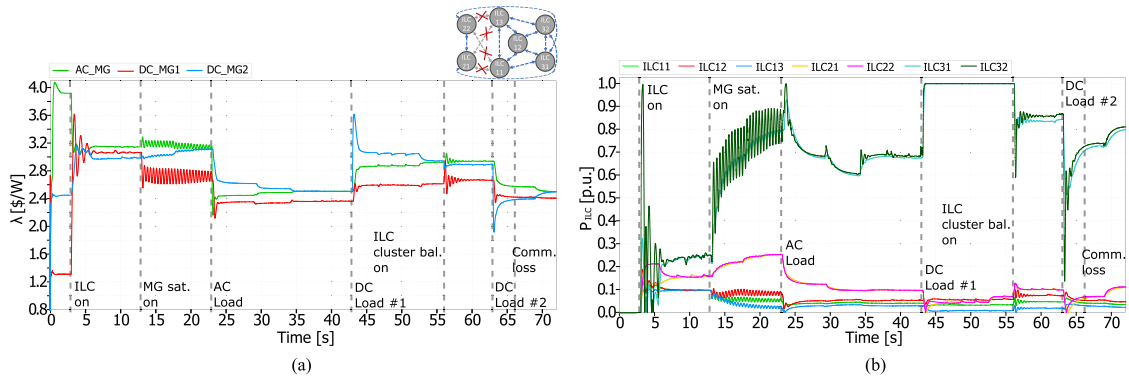

FIGURE 19. Simulation results for Case 4.b. (a) Average IC of MGs. (b) Absolute power of ILCs.

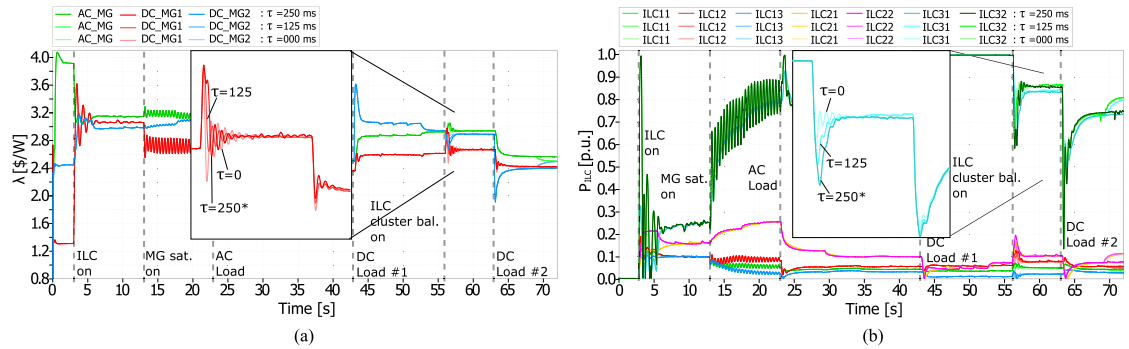

FIGURE 20. Simulation results for Case 4.c. (a) Average IC of MGs. (b) Absolute power of ILCs.

cluster #3, and consequently, to ILC Cluster #2. Also, the relatively slow convergence time of the ILC cluster consensus with respect to the ILC power sharing aids in reducing any transient state overshoot in the face of communication link changes/failures.

L. CASE 4.C. ECONOMIC DISPATCH WITH MG SATURATION CONSTRAINT, ILC POWER SHARING, ILC SATURATION CONSTRAINT, AND COMMUNICATION TIME DELAYS

The results of this test are shown in Fig. 20. In Fig. 20(a), the average ICs present small changes during the transient states. An increase in overshoots can be seen as proportional to the time delay magnitude. Similar behavior is seen for the absolute power of ILCs in Fig. 20(b).

Despite the similarities in the sensibility to time delays of the distributed power observers of (31) and (32), the cluster balancing can afford to face higher delays than the ILC power sharing (see Fig. 15). This can be explained by the fact that the ILC cluster control has a lower control bandwidth and that the communication between ILC clusters is more robust; it does not matter which ILC shares the information since all of them transmit essentially the same ILC average power.

VI. CONCLUSION

This article successfully described a formulation for the economic dispatch in a meshed ac/dc multi-MG with additional objectives. The formulation considered the power constraints of DGs, ILCs, MGs, and clusters of ILCs, fulfilling any critical condition the multi-MG could face. The key factors of the proposed control scheme are the communication network and activation functions (h and Γ parameters).

With this approach, economic dispatch is feasible and robust to obtain since multiple DGs could share the information of IC. Also, the ILCs do not require communication between them at all for the economic dispatch objective. Both aspects are advantages in real-world applications since they reduce the length of communication lines.

For the ILC power sharing, a simultaneous execution can be achieved without altering the dynamics of the economic dispatch, MG saturation loop, and the ILC cluster balancing. However, under time delays, the power observer of the ILC power sharing introduces steady-state delays, which can be dealt with a sophisticated strategy, such as the antiwindup of [14], but at the cost of having chatter in the power waveforms.

The novel balancing of ILC clusters was verified in simulations. This control goal is feasible to implement along with the IC synchronization and the MG power balancing despite its opposition to their control actions to some degree. An essential element for the implementation is the designed weighting factor Γ. If this factor is constant, it always compromises the IC synchronization; therefore, it must be adaptive to the system conditions to be efficient. The full potential of ILC cluster balancing is still under development, and new practical

considerations need to be studied in the multi-MG system to assess all the advantages of giving power reserves to the ILC clusters. Among those considerations are distribution line congestion and power quality compensations, which certainly could provide directives to evaluate the performance and advantages of the control strategy.

Overall, this work opens new lines of research concerning utilizing ILCs for distributed global control goals when system constraints and parameters determined by DNOs are considered.

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